Experimental Stress Analysis

UNIT – I

PRINCIPLES OF EXPERIMENTAL APPROACH

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Merits of Experimental Analysis – Introduction, Uses of Experimental Stress Analysis – Advantages of Experimental Stress Analysis – Different Methods – Simplification of Problems.

1. Introduction to Experimental Stress Analysis (ESA)

- Experimental Stress Analysis = the branch of solid mechanics in which stresses, strains and deformations in a structure or machine part are found by measurement on the real part (prototype) or on a scale model, instead of (or along with) calculation.
- Stress cannot be measured directly. We always measure a strain, a displacement, a force or an optical effect and then convert it to stress using the material's elastic constants (E, ν, G).

$$\sigma = E \cdot \varepsilon \text{ (uniaxial)} \quad \sigma = (E / (1 - \nu^2))(\varepsilon_x + \nu \varepsilon_y) \text{ (biaxial)}$$

- The aim of design: every point of the component should be safe, economical and reliable. For that the designer must know the magnitude, location and direction of the maximum stress.

Why do we need experiments?

- Theory of elasticity gives exact answers only for simple shapes and simple loads (beams, plates, cylinders, discs).
- Real structures have holes, notches, fillets, threads, bolted joints, cut-outs – stress concentration zones where formulae fail.
- Material may be non-homogeneous, anisotropic or non-linear (concrete, composites, soils, rock).
- Boundary conditions, load transfer and support behaviour are uncertain – experiments give the truth, theory gives only an estimate.
- Codes and standards themselves are built on experimental evidence (e.g. cube tests, beam tests).

Brief history (remember 4–5 landmarks)

- 1816 Brewster – discovered photoelasticity (temporary double refraction).
- 1856 Lord Kelvin – showed resistance of a wire changes with strain (basis of strain gauge).
- 1930s Coker and Filon – 2-D photoelasticity became practical.
- 1938 Simmons and Ruge – first bonded wire resistance strain gauge.
- 1950s Foil gauges (printed circuit technique) and brittle-lacquer coatings.

- 1960s onwards – moiré, holography, speckle interferometry, and now digital image correlation (DIC) and fibre-optic sensors.

Background revision (needed for every unit)

- Normal strain $\varepsilon = \Delta L / L$ (change in length per unit length; dimensionless, often quoted in micro-strain, $1\ \mu\varepsilon = 10^{-6}$).
- Shear strain γ = change in the right angle between two lines (radians); Poisson's ratio ν = – lateral strain / axial strain.

$$E = 2G(1 + \nu)\quad K = E / [3(1 - 2\nu)]\quad G = E / [2(1 + \nu)]$$

- Principal stresses (planes of zero shear) for a 2-D state:

$$\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[\,((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2\,]}$$
$$\tan 2\theta = 2\tau_{xy} / (\sigma_x - \sigma_y)\quad \tau_{max} = (\sigma_1 - \sigma_2)/2$$

- Same equations hold for strain with $\sigma \to \varepsilon$ and $\tau \to \gamma/2$ (used in rosette analysis, Unit III).

2. Merits of Experimental Analysis

- Direct and real: result is obtained on the actual material and the actual boundary conditions – no idealisation.
- Works for complex geometry where the mathematical solution is impossible or too lengthy.
- Gives the full-field picture (photoelasticity, brittle coating, moiré) – one look shows the critical zone.
- Checks theory: an experimental result is the standard against which analytical and FEM results are validated.
- Material independent: applicable to metals, plastics, concrete, rock, bone, composites.
- Static as well as dynamic loads (impact, vibration, fatigue, thermal) can be studied.
- Simple models can be tested; small model results are scaled up to prototype by similitude.
- Finds unknown quantities – residual stresses, thermal stresses, contact stresses, assembly stresses – which no designer can calculate easily.

- Safe and economical – failure of a model costs much less than failure of a dam or bridge.
- Builds confidence – used by the designer, the contractor and the court (failure investigation).

Limitations (examiners also ask!)

- Needs skilled operators, costly instruments and a laboratory.
- Each test gives the answer only for that one structure and that loading; parametric studies are slow.
- Model material and scaling errors; gauge installation errors.
- Result is point-wise (strain gauges) or surface only (most optical methods).

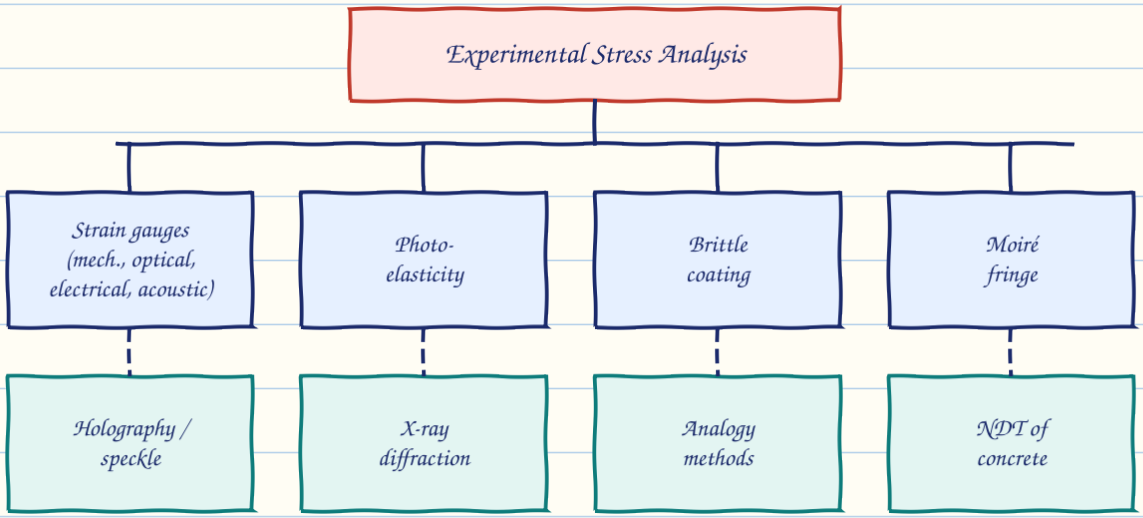


Fig 1.1 Classification of experimental stress analysis methods

3. Uses of Experimental Stress Analysis

- Design verification – proof testing of bridges, dams, buildings, pressure vessels, aircraft and machine frames.
- Finding stress concentration factor (K_t) at holes, notches, keyways, fillets.

$$K_t = \sigma_{max} / \sigma_{nominal}$$

- Locating the critical points so that strain gauges / sensors can be fixed in the right place; also for structural health monitoring.
- Model testing of large structures – dams, shell roofs, chimneys, bridges – before construction (e.g. dam models in laboratories).

- *Residual stress measurement* – welds, castings, rolled sections, shrink fits (hole drilling, X-ray).
- *Measurement of load, torque, pressure, force* (load cells, proving rings, torque meters, pressure transducers use strain gauges!).
- *Material characterisation* – determination of E, ν, ultimate strength, fatigue strength.
- *Failure analysis* – finding why a component cracked or collapsed.
- *Quality control and NDT* – checking concrete strength, cracks, voids (rebound hammer, UPV).
- *Research* – developing new materials, new analytical and numerical methods; validating FEM.

Examples from civil engineering

- Strain gauges on *steel bridge girders* during a load test – compare measured and calculated stresses.
- *Photoelastic model* of a dam / tunnel lining / pile cap to see stress contours.
- *Rebound hammer and UPV* on an old RCC building to assess quality and strength.
- *Vibrating wire gauges* embedded in concrete dams to monitor temperature strains.

4. Advantages of Experimental Stress Analysis

- No need for any *assumptions* about stress distribution – the structure itself tells you.
- *Full-field methods* give stress at every point; *point methods* give accurate numerical values.
- *High accuracy* – modern gauges measure to 1 $\mu\varepsilon$ (10^{-6} strain) or better; error about 1–2 %.
- Usable on *very small* (microscopic) and *very large* (dam) structures.
- *In-service monitoring* is possible – structure is not destroyed (non-destructive).
- Results are *recorded electrically* – easy storage, remote reading, data loggers, computer processing.
- *Fast* for complicated shapes: a photoelastic model can be cut in a few hours.

- Complements numerical methods (FEM, BEM) – both together give a reliable design.

★ Remember: Experimental stress analysis does NOT replace theory – it checks and extends it.

Comparison table (write in answer)

- Analytical method: exact for simple cases only; cheap; fast; many assumptions.
- Numerical (FEM): handles complex geometry; depends on mesh and boundary conditions; needs validation.
- Experimental: real material and real loading; shows true behaviour; needs laboratory and skill; costly for each case.

5. Different Methods of Experimental Stress Analysis

5.1 Strain-gauge methods (Units II and III)

- Measure the change of length / resistance / inductance / frequency over a small gauge length.
- Mechanical (Huggenberger, Berry, dial gauge), optical (Martens mirror, Tuckerman), acoustic (vibrating wire), electrical (inductance, LVDT, capacitance, resistance gauges – most popular).
- Gives strain at a point on the surface; stress from Hooke's law; rosettes give principal strains.

5.2 Photoelasticity (Units IV and V)

- Transparent model of birefringent plastic (epoxy, polycarbonate) is loaded and viewed in polarised light.
- Fringe patterns (isochromatics and isoclinics) give the principal stress difference and direction over the whole model.
- Two-dimensional (transmission), three-dimensional (stress freezing), reflection photoelasticity (birefringent coating on real part).

5.3 Brittle coating method

- Surface of the part is coated with a brittle lacquer (Stresscoat). When the part is loaded, the coat cracks perpendicular to maximum principal tensile strain.

- Crack pattern shows location and direction of critical stress; quick, qualitative; strain level estimated from calibration strip.

5.4 Moiré fringe method

- Two gratings (specimen grating and reference grating) superimposed → moiré fringes that map displacement in the in-plane direction.
- Strain = derivative of displacement: $\varepsilon = (1/p)(\partial N/\partial x)$, p = pitch of grating.

5.5 Holography and speckle interferometry

- Laser light, interference between two states of an object gives displacement to a fraction of the wavelength; good for out-of-plane displacement, vibration, crack detection.

5.6 X-ray diffraction

- Measures change in lattice spacing of metals → residual and applied stresses without touching the part.

5.7 Analogy methods

- Equations of stress problems are similar to those of other phenomena – membrane (soap-film) analogy for torsion of non-circular sections, electrical analogy for plane problems, sand-heap analogy for plastic torsion.

5.8 Non-destructive testing (NDT)

- Rebound hammer, ultrasonic pulse velocity (UPV), radiography, dye penetrant – estimate strength, cracks, voids without damage (Unit III).

5.9 Other modern methods

- Digital Image Correlation (DIC) – camera tracks speckle pattern; gives full-field displacement and strain.
- Fibre-optic (FBG) sensors, thermo-elastic stress analysis (SPATE), acoustic emission.

6. Simplification of Problems

- Real structures are three-dimensional; solving them directly is difficult. The first job of an experimenter is to simplify the problem without losing the essential physics.

6.1 Dimensional reduction

3-D body → 2-D (plane stress / plane strain) → 1-D (bar / beam)

- *Plane stress: thin plate loaded in its own plane.* $\sigma_z = \tau_{xz} = \tau_{yz} = 0$. *Used in 2-D photoelastic models (plate with hole, gear tooth, beam).*
- *Plane strain: very long body with uniform section and load (dam, tunnel, retaining wall, culvert).* $\varepsilon_z = \gamma_{xz} = \gamma_{yz} = 0$ *and* $\sigma_z = \nu(\sigma_x + \sigma_y)$.

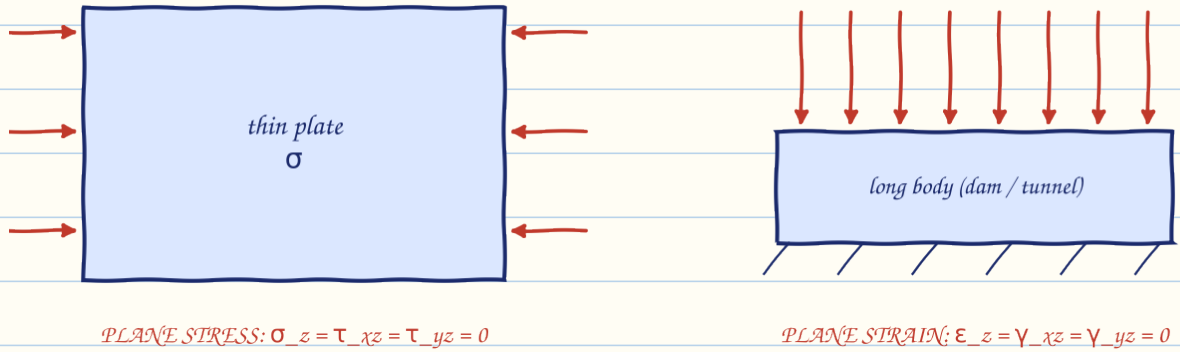


Fig 1.2 Plane stress and plane strain idealisation

6.2 Other simplifications

- *Symmetry: test only one half or one quarter of a symmetric structure with proper boundary conditions.*
- *Saint-Venant's principle: the local effect of a load is felt only up to a distance about equal to the largest dimension of the loaded area; away from it, stresses depend only on the resultant force and moment. So the way a load is applied can be simplified.*
- *Linear elasticity and superposition: if stresses are below the elastic limit, results of separate load cases can be added.*
- *Reduction to a model: use a small-scale, cheaper material model (acrylic, epoxy, micro-concrete) – needs similitude.*
- *Choosing critical sections – measure only where maximum stress is expected (stress concentration zones) rather than everywhere.*
- *Neglect secondary effects (self-weight of small model, friction at supports) if they are small.*

7. Model Theory and Similitude (support for simplification)

- Model and prototype should be geometrically (same shape, all lengths scaled by same factor), kinematically and dynamically similar.

$$\text{Length scale } \lambda_L = L_m / L_p \quad \text{Force scale } \lambda_P = P_m / P_p$$

- For static elastic problems with same Poisson's ratio:

$$\text{Stress scale } \sigma_m / \sigma_n = \lambda_P / \lambda_L^2$$
$$\text{Strain scale } \varepsilon_m / \varepsilon_n = \lambda_P / (\lambda_E \lambda_L^2)$$
$$\text{Deflection scale } \delta_m / \delta_p = \lambda_P / (\lambda_E \lambda_L)$$

- If the model is made of the same material ($\lambda_E = 1$) and loaded so that $\lambda_P = \lambda_L^2$, stresses and strains are equal in model and prototype.
- Buckingham π-theorem is used to form dimensionless groups (e.g. σ/E, $P/(E L^2)$, ν, δ/L) – the groups must be equal for model and prototype.

Example

- A model of a bridge pier is made 1/20 scale, same material. Prototype load = 4000 kN. To get the same stresses, model load $P_m = P_p \lambda_L^2 = 4000 \times (1/20)^2 = 10$ kN.

8. Requirements of a Good Measuring System

- Sensitivity – large output for small strain.
- Range – covers expected strain (elastic strains are only about 10^{-6} to 3×10^{-3}).
- Accuracy and repeatability; linearity and low hysteresis.
- Gauge length short enough to measure local strain (at stress concentration) but long enough to average heterogeneous material (concrete needs long gauges).
- Quick response (frequency response) for dynamic loads.
- Stability – no zero drift; insensitive to temperature, humidity, vibration.
- Light weight, small size (must not stiffen the member), cheap and easy to install.
- Remote reading and recording facilities.

Accuracy, precision, resolution

- Accuracy – closeness of the reading to the true value; precision – closeness of repeated readings to each other; resolution – smallest change the instrument can show (e.g. 1 $\mu\varepsilon$).
- Sensitivity = output change / input change (e.g. mm of pointer travel per micro-strain).

- Types of error: gross (reading/recording mistake), systematic (calibration, zero shift, temperature – can be corrected), random (cannot be avoided – reduced by repeating and averaging).

Sources of error

- Calibration error, gauge-installation error (bad bonding), temperature effect, lead-wire resistance, transverse sensitivity, reinforcing effect of the gauge, human reading error.
- Use statistical treatment: mean $\bar{x} = \Sigma x/n$ and standard deviation $s = \sqrt{[\Sigma (x - \bar{x})^2/(n-1)]}$ to judge the reliability of repeated readings.

9. Steps in an Experimental Investigation

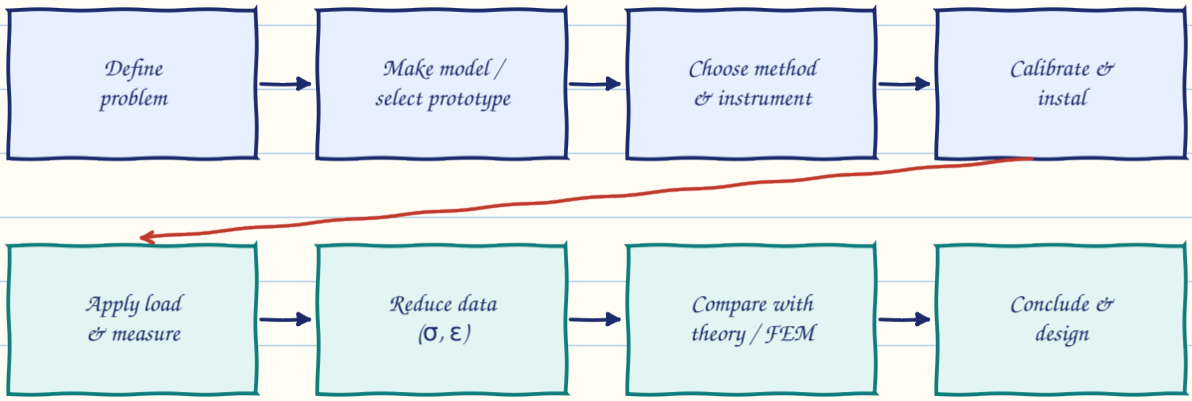


Fig 1.3 Flow chart of an experimental stress analysis

- Define the problem – what quantity ($\sigma, \varepsilon, \delta$) at which location?
- Select the method by considering accuracy, cost, full-field or point, static or dynamic, model or prototype.
- Calibrate the instrument against a known standard.
- Apply loads slowly in steps and record the readings; repeat for reliability.
- Reduce the data – convert to stress using E and ν; plot.
- Compare with theory or FEM, explain differences, give conclusions and design recommendations.

10. Static, Dynamic and Field Load Tests

Static and dynamic measurement

- Static test – load applied slowly; any slow instrument (dial gauge, mechanical gauge, resistance gauge with indicator) can be used.
- Dynamic test – impact, vibration, moving vehicles, earthquakes; needs high frequency response – resistance gauges + oscilloscope / data logger, piezo sensors.
- Gauge used for dynamic work must have natural frequency at least 5–10 times the frequency of the phenomenon.

Procedure of a field load test on a structure (e.g. bridge girder or slab)

- Study the drawings, calculate expected stresses and deflections, decide gauge positions (mid-span, quarter span, near supports).
- Fix strain gauges (bottom and top fibre) and dial gauges / LVDTs for deflection; connect to the data logger and record the zero reading.
- Apply load in increments (25 %, 50 %, 75 %, 100 % of the test load); hold each increment and note readings.
- Hold the full load for the specified time (usually 24 h for a slab, as per IS 456 Cl. 17.6); then unload in stages and check recovery of deflection.
- Compare measured and calculated values; recovery of at least 75 % of the maximum deflection indicates satisfactory behaviour of an RCC member.

★ Linear load–deflection graph and good recovery show that the structure behaved elastically and safely.

11. Solved Numerical Examples

Example 1 – Stress concentration factor

- In a plate with a circular hole the nominal stress is 40 MPa. A gauge at the edge of the hole records a maximum stress of 120 MPa.

$$K_t = \sigma_{max} / \sigma_{nom} = 120 / 40 = 3.0$$

- This agrees with the theoretical value $K_t = 3$ for a small hole in a wide plate (Kirsch solution).

Example 2 – Stress from measured strains (biaxial)

- Steel plate: $E = 200$ GPa, $\nu = 0.3$. Gauges give $\varepsilon_x = 600\ \mu\varepsilon$ and $\varepsilon_y = -100\ \mu\varepsilon$.

$$\sigma_x = E/(1-\nu^2)\,(\varepsilon_x + \nu\,\varepsilon_y) = (200\times10^3/0.91)(600 - 30)\times10^{-6} = 125.3\ \text{MPa}$$

$$\sigma_y = E/(1-\nu^2)\,(\varepsilon_y + \nu\,\varepsilon_x) = (200\times10^3/0.91)(-100 + 180)\times10^{-6} = 17.6\ \text{MPa}$$

- Note: even though ε_y is negative, σ_y is tensile – this is why one gauge alone cannot give stress in a biaxial field.

Example 3 – Model loading

- A 1:10 scale model of a bracket, made of the same material, is to reproduce the prototype stresses. Prototype load = 50 kN.

$$\lambda_L = 1/10, \lambda_E = 1 \rightarrow \lambda_P = \lambda_L^2 = 1/100 \rightarrow P_m = 50/100 = 0.5\ kN$$

- The model deflection is then $\delta_m/\delta_p = \lambda_P/\lambda_L = (1/100)/(1/10) = 0.1$, i.e. 1/10 of prototype deflection.

Example 4 – Plane strain stress

- In a long retaining wall (plane strain) $\sigma_x = 30$ MPa, $\sigma_y = 10$ MPa, $\nu = 0.25$.

$$\sigma_z = \nu(\sigma_x + \sigma_y) = 0.25 \times 40 = 10\ MPa$$

- Although $\varepsilon_z = 0$, a stress σ_z of 10 MPa still exists. In plane stress the reverse is true: $\sigma_z = 0$ but $\varepsilon_z \neq 0$ ($\varepsilon_z = -\nu(\sigma_x + \sigma_y)/E$).

12. Quick Revision – Unit I

- ESA = finding stress and strain by measurement, not calculation.
- Merits: realistic, full-field, handles complex shapes, validates theory.
- Uses: design check, K_t, model tests, residual stress, NDT, failure analysis.
- Methods: strain gauges, photoelasticity, brittle coating, moiré, holography, X-ray, analogies, NDT.
- Simplification: 3-D → 2-D (plane stress / plane strain) → 1-D; symmetry; Saint-Venant; similitude.
- $\sigma_m/\sigma_p = \lambda_P/\lambda_L^2$.

Important questions

- Explain the merits and limitations of experimental stress analysis.
- Write the uses and advantages of ESA in civil engineering.
- Classify the methods of experimental stress analysis and describe any three.
- How are three-dimensional problems simplified? Explain plane stress and plane strain.
- Explain the requirements of a good strain-measuring system.

- State the principle of similitude and derive the stress scale between model and prototype.

References

- Sadhu Singh – Experimental Stress Analysis, Khanna Publishers.
- Dally and Riley – Experimental Stress Analysis, McGraw-Hill.
- L.S. Srinath et al. – Experimental Stress Analysis, Tata McGraw-Hill.

Experimental Stress Analysis

UNIT – II

STRAIN MEASUREMENT USING STRAIN GAUGES

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Syllabus (this unit):

Definition of Strain and its Relation to Experimental Determinations – Properties of Strain Gauge Systems – Types of Strain Gauges: Mechanical, Acoustic and Optical – Introduction to Electrical Strain Gauges – Inductance Strain Gauges – LVDT – Resistance Strain Gauges, various types – Gauge Factor – Materials of Adhesion Base.

1. Definition of Strain and its Relation to Experimental Determination

- Normal (linear) strain = change in length per unit original length.

$$\varepsilon = \Delta L / Lo = (L - Lo) / Lo \text{ (dimensionless; } 1\ \mu\varepsilon = 10^{-6})$$

- Shear strain γ = change of the right angle between two originally perpendicular lines (radians).
- In terms of displacements u, v (small strain theory):

$$\varepsilon_x = \partial u/\partial x \quad \varepsilon_y = \partial v/\partial y \quad \gamma_{xy} = \partial u/\partial y + \partial v/\partial x$$

- Engineering strain ($\Delta L/Lo$) is used for small elastic strains; true strain $\varepsilon_t = \ln(L/Lo)$ is used for large plastic strains. For elastic work both are equal.
- Typical strain levels to remember:
  - Mild steel yield ≈ 0.12 % = 1200 $\mu\varepsilon$ (fy = 250 MPa, $E = 2\times10^5$ MPa)
  - Concrete cracking in tension ≈ 100–150 $\mu\varepsilon$; crushing strain = 3500 $\mu\varepsilon$ (IS 456)
  - Elastic strains in structures are very small, normally 10 – 3000 $\mu\varepsilon$.

Why a special instrument is required

- Take a gauge length Lo = 50 mm and a strain of 1 $\mu\varepsilon$. The change in length is

$$\Delta L = \varepsilon\ Lo = 10^{-6} \times 50\ mm = 0.00005\ mm\ (= 0.05\ \mu m)$$

- Such a small deformation can only be read after magnification – mechanical (levers), optical (light beam), or electrical (change of resistance, inductance, frequency).

Relation to experimental stress determination

- Stress is never measured directly. We measure strain over the gauge length (average strain) and compute stress by Hooke's law.

Uniaxial: $\sigma = E\,\varepsilon$ Biaxial (free surface, plane stress): $\sigma_x = E\,(\varepsilon_x + \nu\,\varepsilon_y)/(1 - \nu^2)$

- For 3-D: $\varepsilon_x = [\sigma_x - \nu(\sigma_y + \sigma_z)]/E$ and so on.
- Strain gauges measure only on the surface, which is the free surface where $\sigma_z = 0$ – so plane-stress equations are valid.

- If principal directions are known (beam, pressure vessel, bar) 1–2 gauges are enough; if unknown, three-element rosettes are needed (Unit III).

2. Properties (Characteristics) of Strain Gauge Systems

- Gauge length (Lo) – the original length over which the strain is averaged. Short gauge → local strain (stress concentration); long gauge → averages (concrete).
  - Metals: 3 – 10 mm, concentration zones: 0.2 – 1.5 mm; concrete: 50 – 150 mm (at least 3–4 times the maximum aggregate size).
- Gauge sensitivity – output per unit strain (e.g. mm/με, Ω/με, mV/V per με).
- Range – largest strain that can be read (resistance gauges 0.5–5 %, semiconductor ≤ 0.3 %).
- Accuracy and resolution – smallest strain change observable (1 με for good electrical systems).
- Linearity and hysteresis – output must be proportional to strain; same reading for loading and unloading.
- Frequency response – mechanical gauges are static only; electrical gauges respond to 100 kHz or more.
- Stability – no zero drift with time, temperature, humidity.
- Size and weight – should not reinforce the member or load it (inertia in dynamic tests).
- Ease of installation, remote reading, low cost, reusability.

Ideal strain gauge: small, light, cheap, high sensitivity, linear, fast, stable, temperature-insensitive, remote readable.

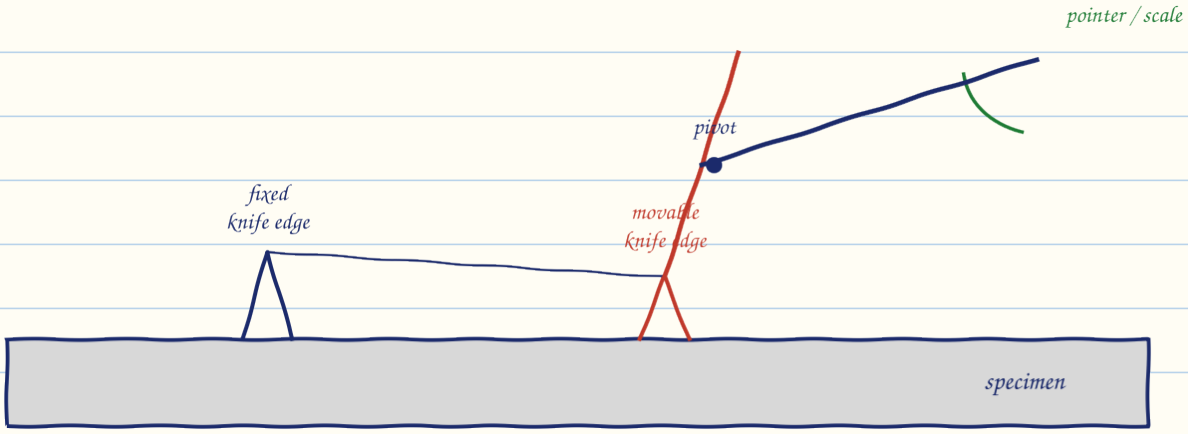
Classification of strain gauges

- Mechanical → Optical → Acoustic → Electrical (inductance, capacitance, resistance, piezo-electric).

3. Mechanical Strain Gauges

- Strain is magnified by levers, gears or dial-gauge mechanism. Gauge is held on the specimen by knife edges / conical studs.
- Examples:

- Huggenberger tensometer – compound lever, magnification 300 to 2000, gauge length 12 – 25 mm.
- Berry strain gauge – dial gauge on a bar (gauge length 200 mm), used on concrete and masonry.
- DEMEC (demountable mechanical) gauge – studs fixed on the surface, gauge lengths 100 / 200 mm, digital dial; accuracy ≈ 5 με.
- Johansson Mikrokator – twisted metal strip magnifies motion (100 – 5000 times).



Huggenberger tensometer: δ at gauge length L → pointer travel = (a/b)·δ

Fig 2.1 Huggenberger tensometer (compound lever)

- Magnification of the lever system:

$$M = (a / b)\cdot(c / d) = \text{pointer travel} / \text{change in length}$$

- Advantages: simple, cheap, needs no power, can be reused, no calibration drift.
- Limitations: friction and backlash errors, large size, only static strain, accessible surface needed, accuracy is low (about 10–20 με), tedious reading.

4. Optical Strain Gauges

- Use the reflection of light from a small rotating mirror – a weightless pointer which gives very high magnification, no inertia or friction.

Martens mirror (optical lever) gauge

- A knife edge is fixed on the specimen; a second V-shaped edge turns a mirror by an angle θ when the specimen deforms. A telescope and scale at distance D reads the movement.

$\theta = \delta / d$ (δ = change in gauge length, d = distance between knife edges)

Scale deflection $s = 2D\theta = 2D\delta / d \rightarrow$ Magnification $= 2D / d$

- Example: $d = 10$ mm, $D = 1000$ mm → magnification = 200. With $\delta = 0.005$ mm the scale moves 1 mm.

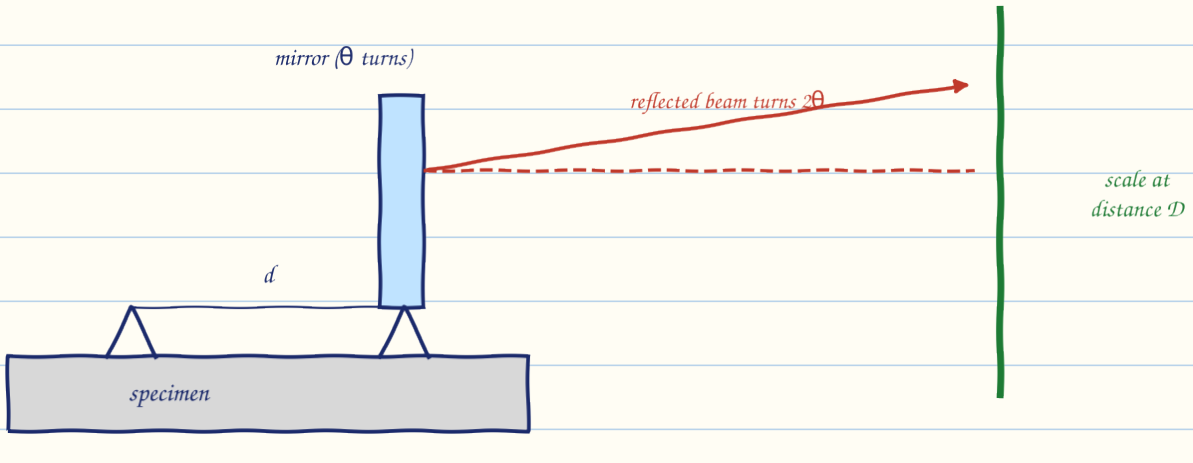


Fig 2.2 Martens mirror optical strain gauge

Tuckerman optical gauge

- Uses an autocollimator and a lozenge (rhombic) attached to the specimen; sensitivity 2×10^{-6} strain, the most accurate mechanical-optical gauge; used mainly for calibration of other gauges in the laboratory.
- Advantages: high sensitivity, no inertia, remote reading, high accuracy.
- Limitations: delicate, expensive, bulky, needs steady table and dim light; only static strains; difficult on site.

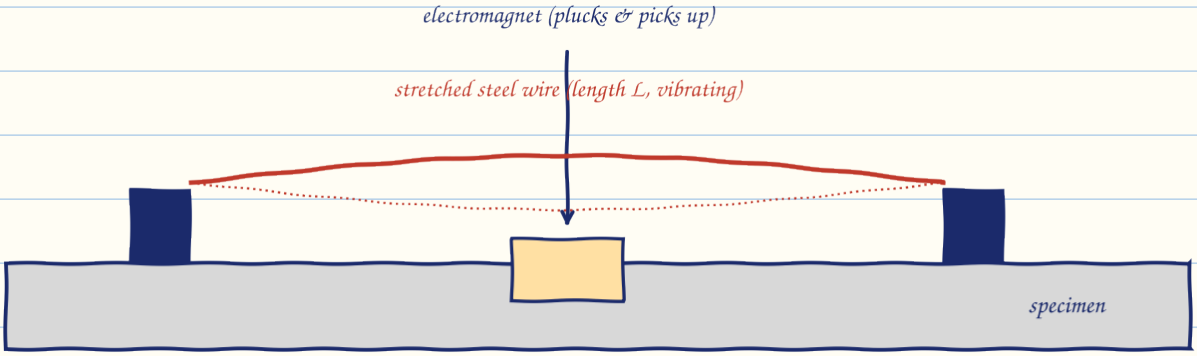
5. Acoustic Strain Gauges (Vibrating Wire Gauge)

- A steel wire is stretched between two end blocks fixed to the member. When the member strains, the wire tension and hence its natural frequency changes.
- Frequency of a stretched wire (length L, tension T, mass per unit length m):

$$f = (1 / 2L)\sqrt{(T / m)}$$

- With $T = \sigma_w A$ and $m = \rho A$: $\sigma_w = 4 \rho L^2 f^2$, so the strain is proportional to f^2.

$$\Delta\varepsilon = (4 \rho L^2 / E_w)(f^2 - f_0^2) = K(f^2 - f_0^2)$$



$f = (1/2L)\cdot\sqrt{(T/m)} \rightarrow \text{strain} \propto (f^2 - f_0^2)$

Fig 2.3 Vibrating-wire (acoustic) strain gauge

- An electromagnet plucks the wire and then picks up its frequency, which is read by a frequency counter.
- Example: steel wire $\rho = 7850\ kg/m^3$, $E = 2\times10^{11}\ Pa$, $L = 0.1\ m$; $f_0 = 800\ Hz$, $f = 1000\ Hz$.

$$\Delta\varepsilon = (4 \times 7850 \times 0.1^2/2\times10^{11})\,(10^6 - 0.64\times10^6) = 5.65\times10^{-4} = 565\ \mu\varepsilon$$

- Advantages: very stable for years, signal is frequency (not affected by cable length, contact resistance, moisture) – ideal embedded in concrete dams, tunnel linings, foundations; built-in thermistor for temperature.
- Limitations: static or slowly varying strain only (the wire needs time to vibrate), larger size, higher cost.

6. Electrical Strain Gauges – Introduction

- Strain is converted into an electrical signal (resistance, inductance or capacitance change), which is amplified and displayed/recorded.
- Advantages over mechanical/optical: high sensitivity, small size and weight, low cost per point, remote and multi-point reading, static and dynamic measurement, data logging, high accuracy, can be used under water, at high or low temperature, embedded in structures.

Types

- Inductance type (including LVDT), capacitance type, resistance type (most popular), piezo-electric, semiconductor.

6.1 Inductance strain gauge

- A coil with an iron armature; strain changes the air gap or the armature position, hence the inductance L:

  $L = N^2 / R_m$ with reluctance $R_m = \ell / (\mu A)$ (N turns, ℓ air gap, A area)

- The change of inductance is measured with an AC bridge (carrier frequency 1–20 kHz). Variable air gap, variable area and variable core-position types exist.

6.2 LVDT – Linear Variable Differential Transformer

- Construction: one primary coil at the centre, two identical secondary coils on both sides (connected in series opposition) and a movable soft-iron core attached to the member whose displacement is to be measured.

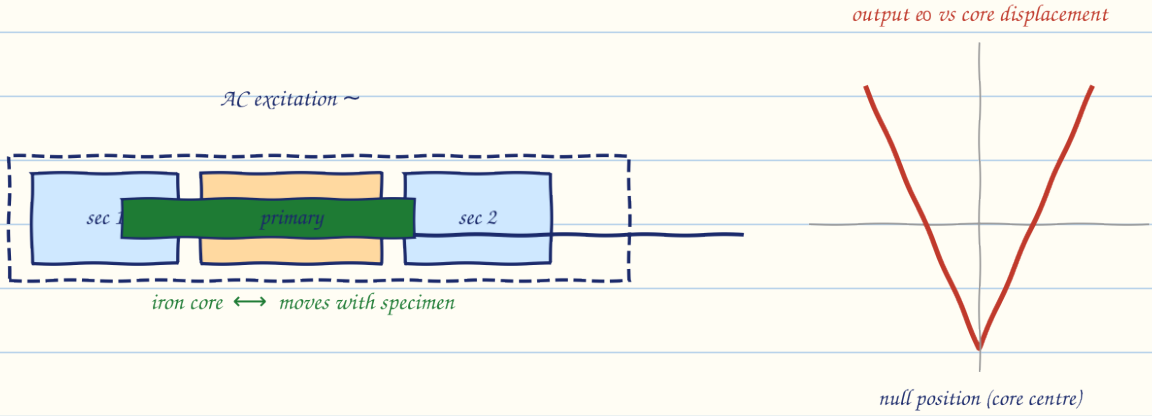


Fig 2.4 LVDT and its output characteristic

- Working: AC excitation (e.g. 5 V, 2.5 kHz) is given to the primary.
  - Core at centre → equal emf in both secondaries, opposite sign → net output = 0 (null position).
  - Core moves up → emf in one secondary rises, in other falls → output proportional to displacement; the phase (0° or 180°) shows the direction.

  $e_0 = e_{s1} - e_{s2} \propto x$ (linear over ±(1 to 250) mm depending on size)

- Advantages: no friction or wear, very high resolution, high sensitivity, robust, small hysteresis, infinite life, can measure deflections, settlement, crack opening, load (with proving rings).
- Limitations: needs AC supply + demodulator, core mass limits frequency (signal frequency should be below 1/10 of carrier), affected by stray magnetic fields;

linear only for the central region.

6.3 Capacitance gauge (brief)

- Capacitance $C = \varepsilon A / d$ changes when the plate gap d or area A changes with strain; very sensitive and good at high temperature, but needs shielded cables and careful electronics.

7. Electrical Resistance Strain Gauge (ERSG)

7.1 Principle

- Resistance of a conductor of resistivity ρ, length L and area A:

$$R = \rho L / A$$

- Taking logarithms and differentiating:

$$dR/R = d\rho/\rho + dL/L - dA/A$$

- With $A = \pi d^2/4$, $dA/A = 2\, dd/d = -2\nu\varepsilon$, so

$$\Delta R/R = \varepsilon\,(1 + 2\nu) + \Delta\rho/\rho$$

- $(1 + 2\nu)$ = geometry effect (length increases, area decreases); $\Delta\rho/\rho$ = piezo-resistive effect (resistivity changes with strain).

7.2 Gauge factor (strain sensitivity)

$$S_g = (\Delta R/R)/\varepsilon = 1 + 2\nu + (\Delta\rho/\rho)/\varepsilon$$

- For most metal alloys $S_g \approx 2.0 – 2.1$; for semiconductor gauges 100 to 175 (positive for p-type, negative for n-type); for nickel $S_g \approx -12$.
- Example: gauge $R = 120\ \Omega$; at 1000 $\mu\varepsilon$ the resistance rises by 0.24 Ω.

$S_g = (0.24/120)/(1000\times10^{-6}) = 2.0$ and for $S_g = 2.1$: $\Delta R = 120 \times 2.1 \times 10^{-3} = 0.252\ \Omega$

- Note how small ΔR is (0.2 %), so a Wheatstone bridge is needed (Section 9).

7.3 Requirements of the gauge material

- High gauge factor, high resistivity, low temperature coefficient of resistance, good linearity, high yield strain, good fatigue life, easy to roll into foil / draw into wire, low thermo-emf against copper.

7.4 Common gauge alloys

- Constantan (Advance, Cu 55 / Ni 45) – $S_g = 2.1$, most popular, self temperature-compensated, up to 10 % strain (static), 300 °C.

- Karma (Ni-Cr alloy) – S_g = 2.0; excellent fatigue life and stability, – 270 to 290 °C.
- Isoelastic (Ni-Fe-Cr-Mo) – S_g = 3.5; good for dynamic tests (high output), poor temperature stability.
- Platinum-tungsten – S_g = 4.0; high-temperature work (up to 800 °C).
- Silicon / germanium – S_g about 100–175 but non-linear and very temperature sensitive.

8. Types of Resistance Strain Gauges (Various Types)

- 1. Unbonded wire gauge – wire stretched over insulated pins (Statham gauge); used in transducers (load cells, pressure cells), not for direct fixing.
- 2. Bonded wire gauge – a fine wire (0.025 mm dia.) in a flat grid, glued between paper layers; patterns: flat grid, wrap-around, helical. Early type; now replaced by foil.
- 3. Bonded metal foil gauge (most common) – grid photo-etched from 3–5 µm foil on a plastic backing; solder tabs.
  - Advantages: large heat-dissipation area (higher bridge voltage), good fatigue life, any grid pattern can be etched, stable, cheap, easily bonded.

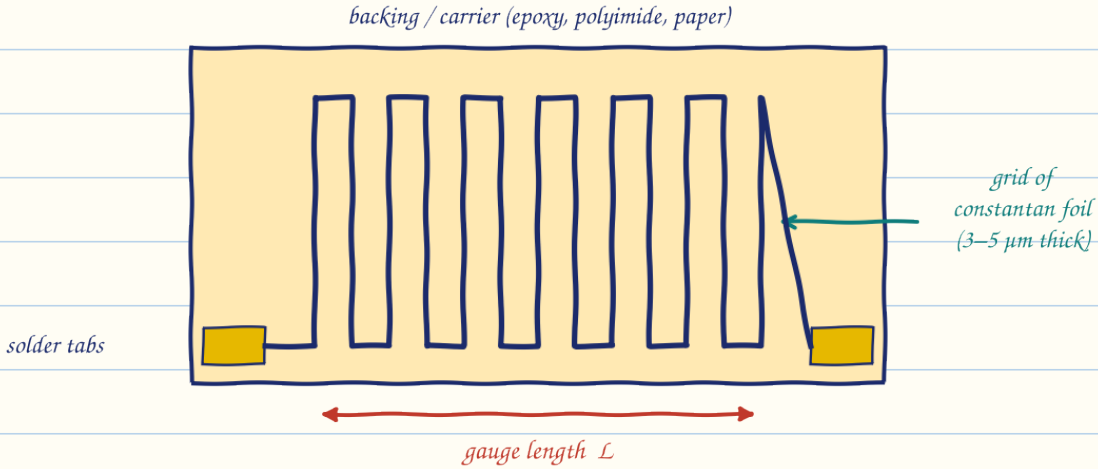


Fig 2.5 Bonded metal foil strain gauge

- 4. Semiconductor gauge – piezoresistive silicon strip; S_g = 100–175; used for very small strains and in miniature transducers; limitations: non-linear, temperature

sensitive, brittle.

- 5. Weldable gauges – foil gauge on a thin steel strip spot-welded to steel structures (bridges, ships, field work).
- 6. Embedment gauges for concrete – 60–150 mm long gauge in a protective casing, cast in concrete.
- 7. Thin film and capacitance gauges – for high temperature and very high-frequency work.

Gauge patterns

- Uniaxial (single grid), biaxial (tee, 0°/90°), rosettes (rectangular 0-45-90, delta 0-60-120), shear gauges (±45°), diaphragm and bending-beam patterns used in transducers.
- Standard gauge resistance: 120 Ω, 350 Ω, 1000 Ω; gauge length 0.2 mm – 100 mm.
- Self-temperature-compensated (STC) gauges are matched to the thermal expansion of the test material (steel, aluminium).

9. Materials of Adhesion Base (Backing / Carrier and Adhesive)

9.1 Backing (carrier) material

- Holds the grid, insulates it electrically and transfers strain from the structure to the grid. It should be thin, flexible, strong, moisture resistant, and compatible with the adhesive.
  - Paper (thin bakelite-impregnated) – cheap, for general static work, up to about 80 °C; absorbs moisture.
  - Epoxy / bakelite film – better stability, – 30 to 100 °C.
  - Polyimide (Kapton) – flexible, high elongation, – 270 to 290 °C; very common.
  - Glass-fibre-reinforced epoxy-phenolic – for high temperature (to 290 °C) and good fatigue.
  - Mica / ceramic / metal carriers – very high temperatures.

9.2 Adhesives (cement)

- Must give a thin, uniform, strong shear bond, cure at room temperature or with mild heat, be electrically insulating and non-creeping.
  - Nitrocellulose (cellulose nitrate) cement – air drying, general purpose, quick; moderate temperature.
  - Cyanoacrylate (instant adhesive) – cures in 1 min, convenient for field and short term tests.
  - Epoxy resin (Araldite) – best for long-term, high accuracy, large elongation, –270 to 100 °C (heat-cured to 200 °C); preferred for concrete (after surface levelling).
  - Polyester / phenolic – moderate temperature.
  - Ceramic cements (Rokide, Sauereisen) – for very high temperature (above 500 °C).

9.3 Installation procedure (important)

- Surface preparation: degrease (solvent) → abrade with 220–320 grit paper → clean with conditioner → neutralise.
- Mark gauge axis with a pencil / scribe; apply a thin film of adhesive; place gauge; press with thumb through Teflon tape for 1–2 min; allow curing.
- Solder lead wires to the tabs (using terminals to relieve strain); check insulation resistance (> 1000 MΩ) and gauge resistance.
- Protect the gauge with wax, silicone rubber or epoxy coating against moisture and damage.
- On concrete, first seal the surface with epoxy filler / putty, level it, then bond the gauge.

10. Wheatstone Bridge Circuit for Strain Gauges

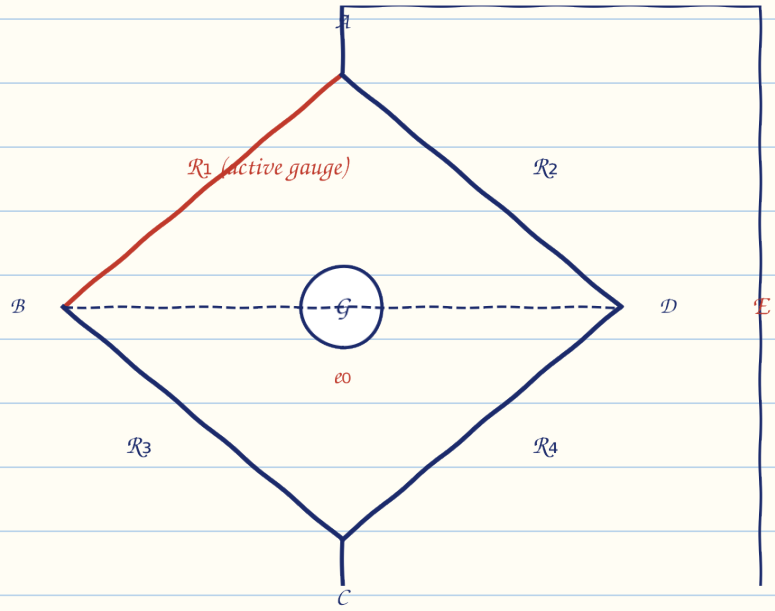


Fig 2.6 Wheatstone bridge with an active gauge R1

- DC excitation E is given across A–C; the output eo is taken across B–D. For the sketch (arms R1 R2 R4 R3 taken round the loop):

$$e_o = E\,(R_2R_3 - R_1R_4)\,/\,[(R_1 + R_3)(R_2 + R_4)]$$ balanced bridge: $R_1R_4 = R_2R_3 \rightarrow e_o = 0$

- If every arm is R and the resistance changes are small:

$$e_o = (E/4)(\Delta R_1/R_1 - \Delta R_2/R_2 + \Delta R_3/R_3 - \Delta R_4/R_4) = (E\,S_g/4)(\varepsilon_1 - \varepsilon_2 + \varepsilon_3 - \varepsilon_4)$$

- Rule: strains in opposite arms add; in adjacent arms they subtract (numbered round the bridge).

Bridge arrangements

- Quarter bridge (one active gauge + 3 fixed resistors, preferably a dummy gauge in the adjacent arm for temperature compensation):

$$e_o = E\,S_g\,\varepsilon\,/\,4$$

- Half bridge (two active gauges, adjacent arms – bending: one in tension, one in compression):

$$e_o = E\,S_g\,\varepsilon\,/\,2$$ (Poisson-gauge pair: $e_o = E\,S_g\,\varepsilon\,(1 + \nu)/4$)

- Full bridge (four active gauges; bending with 2 tension + 2 compression; load cells):

$$e_o = E\,S_g\,\varepsilon$$ (axial load with Poisson gauges: $e_o = E\,S_g\,\varepsilon\,(1 + \nu)/2$)

- Example: $R = 120\ \Omega$, $S_g = 2.1$, $\varepsilon = 1000\ \mu\varepsilon$, $E = 5\ V$:

  Quarter: $e_o = 5 \times 2.1 \times 10^{-3}/4 = 2.63\ mV$ Half: 5.25 mV Full: 10.5 mV
- Temperature compensation: the dummy gauge (same lot, same material, unstressed, same temperature) in the adjacent arm cancels thermal resistance change automatically.
- Calibration by shunt resistor R_c across one arm: equivalent strain $\varepsilon \approx R/(S_g(R + R_c))$; $R_c = 60\ k\Omega$, $R = 120\ \Omega$, $S_g = 2.0 \rightarrow \varepsilon \approx 997\ \mu\varepsilon$ ($\approx 1000\ \mu\varepsilon$).
- Lead-wire resistance and contact resistance cause error in quarter bridge – use 3-wire connection.
- Readout: static strain indicator (null balance), data logger, dynamic amplifier + CRO.

11. Quick Revision – Unit II

- $\varepsilon = \Delta L/L_o$; stress from Hooke's law; very small strains need magnification.
- Gauge properties: gauge length, sensitivity, range, accuracy, response, stability, size.
- Mechanical (Huggenberger, Berry, DEMEC), Optical (Martens mirror: $M = 2D/d$; Tuckerman), Acoustic ($f = (1/2L)\sqrt{(T/m)}$, $\varepsilon \propto f^2$).
- Electrical: inductance, LVDT (primary + 2 secondaries, null position, ± displacement), resistance gauge.

$S_g = (\Delta R/R)/\varepsilon = 1 + 2\nu + (\Delta\rho/\rho)/\varepsilon \approx 2$ for metals; 100–175 for semiconductor.

- Foil gauge best; constantan S_g 2.1; backing: paper, epoxy, polyimide; adhesives: nitrocellulose, cyanoacrylate, epoxy, ceramic.
- Bridge: $e_o = E S_g \varepsilon/4$ (quarter), /2 (half), full: $E S_g \varepsilon$.

Important questions

- Define strain; explain how stress is obtained from strain measurements.
- List the requirements and properties of a good strain gauge system.
- Describe the working of Huggenberger tensometer and Martens mirror gauge.
- Explain the vibrating-wire acoustic gauge and derive the relation between frequency and strain.
- Explain the principle of LVDT. List its advantages and applications.

- Define gauge factor, derive $S_g = 1 + 2\nu + (\Delta\rho/\rho)/\varepsilon$.
- Discuss types of resistance strain gauges, the materials used and the adhesives and backing.
- Explain quarter, half and full bridge circuits; why is a dummy gauge used?

Experimental Stress Analysis

UNIT – III

STRAIN ROSETTES AND NON-DESTRUCTIVE TESTING OF CONCRETE

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Introduction – The Three-Element Rectangular Rosette – The Delta Rosette – Corrections for Transverse Strain Gauge – Ultrasonic Pulse Velocity Method – Application to Concrete – Hammer Test – Application to Concrete.

Handwritten-style study notes · B.Tech Civil Engineering – IV Year

1. Introduction – Why Strain Rosettes?

- A single strain gauge measures strain only along its own axis.
- At a point on a free surface (plane stress) there are three unknowns:

$$\varepsilon_x, \varepsilon_y, \gamma_{xy} \text{ or equivalently } \varepsilon_1, \varepsilon_2, \theta \text{ (principal strains and direction)}$$

- Hence three independent strain readings in three different directions are needed. A group of three (or four) gauges at one point is called a strain rosette.
- If the principal directions are known (simply supported beam, bar in tension, pressure vessel) – two gauges (0° and 90°) or even one gauge is enough.
- If the directions are unknown (joints, plates with holes, bracket, around openings) – a rosette is a must.

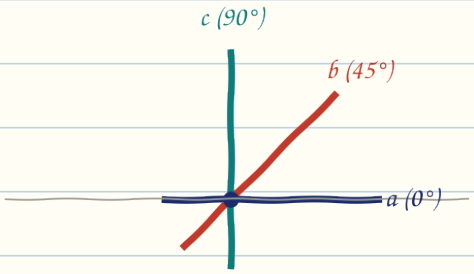
Strain at any angle θ (transformation equation)

$$\varepsilon_\theta = (\varepsilon_x + \varepsilon_y)/2 + (\varepsilon_x - \varepsilon_y)/2 \cdot \cos 2\theta + (\gamma_{xy}/2) \cdot \sin 2\theta$$

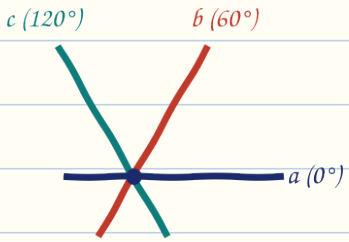
- Writing this for three gauge angles $\theta_a, \theta_b, \theta_c$ gives three equations for the three unknowns.

Common rosette types

- Rectangular (0°-45°-90°), Delta (0°-60°-120°), Tee-delta, and four-element 0-45-90-135 (extra gauge gives a check $\varepsilon_a + \varepsilon_c = \varepsilon_b + \varepsilon_d$).
- Built as planar (all grids on one foil – cheap, small) or stacked (grids one above another – measure at one exact point; poorer heat dissipation).



Rectangular rosette (0°, 45°, 90°)



Delta rosette (0°, 60°, 120°)

Fig 3.1 Rectangular and delta rosettes

2. Three-Element Rectangular Rosette

- Gauges a, b, c at 0°, 45° and 90° to the reference x-axis; readings $\varepsilon_a, \varepsilon_b, \varepsilon_c$.
- Substituting $\theta = 0°, 45°, 90°$ in the transformation equation:

$$\varepsilon_a = \varepsilon_x \quad \varepsilon_c = \varepsilon_y \quad \varepsilon_b = (\varepsilon_x + \varepsilon_y)/2 + \gamma_{xy}/2$$
$$\therefore \gamma_{xy} = 2\varepsilon_b - \varepsilon_a - \varepsilon_c$$

Principal strains and direction

$$\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_c)/2 \pm (1/\sqrt{2})\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2]}$$
$$\tan 2\theta_n = (2\varepsilon_b - \varepsilon_a - \varepsilon_c)/(\varepsilon_a - \varepsilon_c) \quad (\theta_n \text{ measured from gauge a})$$
$$\gamma_{max} = \varepsilon_1 - \varepsilon_2 = \sqrt{2}\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2]}$$

Principal stresses (Hooke's law, plane stress)

$$\sigma_{1,2} = (E/2)\{(\varepsilon_a + \varepsilon_c)/(1-\nu) \pm [\sqrt{2}/(1+\nu)]\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2]}\}$$
$$\tau_{max} = (\sigma_1 - \sigma_2)/2 = E(\varepsilon_1 - \varepsilon_2)/[2(1+\nu)]$$

- Mohr's circle of strain gives the same results graphically: plot $(\varepsilon_x, \gamma_{xy}/2)$ and $(\varepsilon_y, -\gamma_{xy}/2)$; centre at $(\varepsilon_x + \varepsilon_y)/2$.

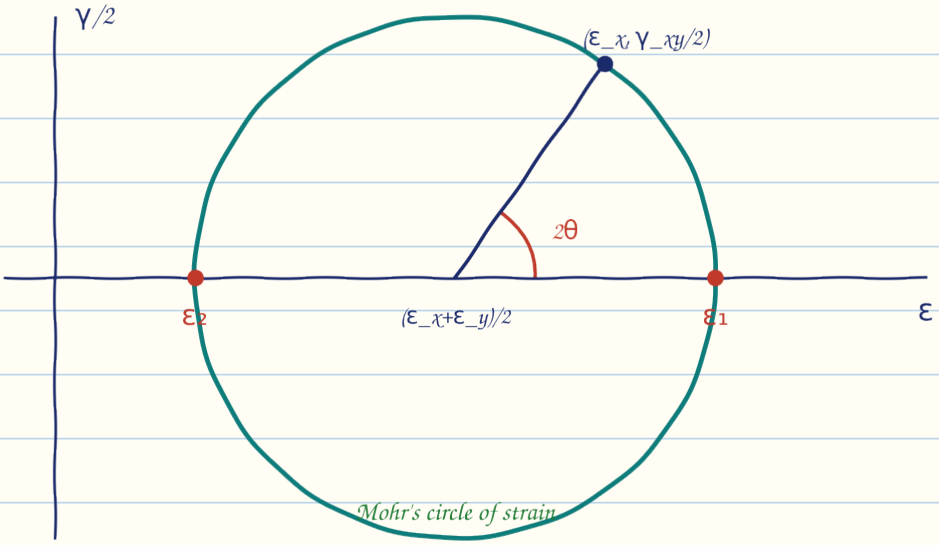


Fig 3.2 Mohr's circle of strain for rosette readings

Steps to construct Mohr's circle of strain from rosette readings

- First find $\varepsilon_x = \varepsilon_a$, $\varepsilon_y = \varepsilon_c$ and $\gamma_{xy} = 2\varepsilon_b - \varepsilon_a - \varepsilon_c$.
- Draw axes: ε horizontal, $\gamma/2$ vertical (clockwise shear taken positive downward by the usual convention).
- Plot points X $(\varepsilon_x, \gamma_{xy}/2)$ and Y $(\varepsilon_y, -\gamma_{xy}/2)$; join them – the line cuts the ε-axis at the centre $C = (\varepsilon_x + \varepsilon_y)/2$.

- Draw the circle with centre C and radius $R = XC$. Its intercepts on the ε-axis give ε_1 (right) and ε_2 (left); top of circle gives $\gamma_{max}/2 = R$.
- The angle XC to the ε-axis is $2\theta_p$; the principal axis is rotated by θ_p from gauge a (in the same sense as the circle angle).

Check with a known field (uniaxial tension)

- Let the rosette lie along a bar with uniaxial stress σ in direction a: $\varepsilon_a = \varepsilon = \sigma/E$, $\varepsilon_c = -\nu\varepsilon$, and since $\gamma_{xy} = 0$, $\varepsilon_b = (\varepsilon_a + \varepsilon_c)/2 = \varepsilon(1 - \nu)/2$.

$$\varepsilon_1 = \varepsilon(1 - \nu)/2 + (1/\sqrt{2})\,\varepsilon\,(1 + \nu)/2 \cdot \sqrt{2} = \varepsilon(1 - \nu)/2 + \varepsilon(1 + \nu)/2 = \varepsilon \text{ and } \theta_p = 0°$$

- So the formulae return the true strain and the correct axis – a useful check on a new rosette installation.

Practical points for rosette installation

- Align the a-gauge with the reference axis (scribe line) within ±1°; an angular error of 1° gives a noticeable error in θ_p and shear.
- Connect each element as a quarter bridge with its own dummy (or a common dummy) and use the same lead length for all three.
- Keep the zero reading before loading and record all three channels at the same load step.

Solved example 1

- Rectangular rosette on steel ($E = 200$ GPa, $\nu = 0.3$): $\varepsilon_a = 800$, $\varepsilon_b = 400$, $\varepsilon_c = -200\ \mu\varepsilon$.

$$\text{Centre} = (800 - 200)/2 = 300\ \mu\varepsilon;\ (\varepsilon_a - \varepsilon_b) = 400,\ (\varepsilon_b - \varepsilon_c) = 600$$

$$\text{radius} = (1/\sqrt{2})\sqrt{(400^2 + 600^2)} = 509.9\ \mu\varepsilon$$

$$\varepsilon_1 = 809.9\ \mu\varepsilon,\ \varepsilon_2 = -209.9\ \mu\varepsilon$$

$$\tan 2\theta_p = (800 - 800 + 200)/(800 + 200) = 0.2 \rightarrow 2\theta_p = 11.3°,\ \theta_p = 5.65° \text{ from gauge a}$$

$$\sigma_1 = E/(1 - \nu^2)\,(\varepsilon_1 + \nu\,\varepsilon_2) = (200\times10^3/0.91)(809.9 - 63.0)\times10^{-6} = 164.2\ \text{MPa}$$

$$\sigma_2 = (200\times10^3/0.91)(-209.9 + 243.0)\times10^{-6} = 7.3\ \text{MPa}$$

3. The Delta Rosette (0°-60°-120°)

- Gauges a, b, c at 0°, 60°, 120° (equiangular – 60° apart).

$$\varepsilon_a = \varepsilon_x \quad \varepsilon_b = \varepsilon_x/4 + 3\varepsilon_y/4 + (\sqrt{3}/4)\gamma_{xy} \quad \varepsilon_c = \varepsilon_x/4 + 3\varepsilon_y/4 - (\sqrt{3}/4)\gamma_{xy}$$

- Solving:

$$\varepsilon_x = \varepsilon_a \quad \varepsilon_y = [2(\varepsilon_b + \varepsilon_c) - \varepsilon_a] / 3 \quad \gamma_{xy} = 2(\varepsilon_b - \varepsilon_c)/\sqrt{3}$$

Principal strains, directions and stresses

$$\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_b + \varepsilon_c)/3 \pm (\sqrt{2}/3)\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2 + (\varepsilon_c - \varepsilon_a)^2]}$$

$$\tan 2\theta_n = \sqrt{3}(\varepsilon_b - \varepsilon_c) / (2\varepsilon_a - \varepsilon_b - \varepsilon_c)$$

$$\sigma_{1,2} = E\{(\varepsilon_a + \varepsilon_b + \varepsilon_c)/[3(1-\nu)] \pm [\sqrt{2}/(3(1+\nu))]\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2 + (\varepsilon_c - \varepsilon_a)^2]}\}$$

Solved example 2

- Delta rosette on steel ($E = 200$ GPa, $\nu = 0.3$): $\varepsilon_a = 600$, $\varepsilon_b = 300$, $\varepsilon_c = -100$ $\mu\varepsilon$.

$(\varepsilon_a + \varepsilon_b + \varepsilon_c)/3 = 266.7\ \mu\varepsilon$; $\Sigma(\text{difference})^2 = 300^2 + 400^2 + 700^2 = 740\,000 \rightarrow \sqrt{} = 860.2$

radius $= (\sqrt{2}/3)(860.2) = 405.5\ \mu\varepsilon \rightarrow \varepsilon_1 = 672.2\ \mu\varepsilon$, $\varepsilon_2 = -138.8\ \mu\varepsilon$

$\tan 2\theta_n = \sqrt{3}(400)/(1200 - 300 + 100) = 0.693 \rightarrow \theta_n = 17.4°$ from gauge a

$\sigma_1 = (200\times10^3/0.91)(672.2 - 41.6)\times10^{-6} = 138.6$ MPa; $\sigma_2 = (200\times10^3/0.91)(-138.8 + 201.7)\times10^{-6} = 13.8$ MPa

Rectangular vs delta

- Rectangular: simpler formulae, ε_x and ε_y read directly; the most used.
- Delta: all gauges equally spaced (60°), better uniform accuracy in every direction and more compact; formulae slightly longer.
- Both give identical results if the readings correspond to the same strain field.

4. Corrections for Transverse Strain (Transverse Sensitivity)

- A strain gauge grid has end loops (turn-around) and a finite width, so it responds slightly to the strain perpendicular to its axis.

$$\Delta R/R = S_a(\varepsilon_a + K_t\varepsilon_t) \quad K_t = S_t/S_a \text{ (transverse sensitivity factor)}$$

- S_a = axial sensitivity, S_t = transverse sensitivity; K_t is typically 0 – 5 % (foil gauges ≈ 0 – 2 %, wire-wrap gauges higher).
- Manufacturer's gauge factor S_g is measured on a beam of Poisson's ratio $\nu_0 = 0.285$, where $\varepsilon_t = -\nu_0\varepsilon_a$, so

$$S_g = S_a (1 - \nu_o K_t)$$

- Indicated strain (using S_g): $\varepsilon' = (\varepsilon_a + K_t \varepsilon_t) / (1 - \nu_o K_t)$. True strains for a 0°/90° (two-gauge) arrangement:

$$\varepsilon_a = [(1 - \nu_o K_t)/(1 - K_t^2)] (\varepsilon'_a - K_t \varepsilon'_c)$$
$$\varepsilon_c = [(1 - \nu_o K_t)/(1 - K_t^2)] (\varepsilon'_c - K_t \varepsilon'_a)$$

- For the 45° gauge of a rectangular rosette (transverse strain = $\varepsilon_a + \varepsilon_c - \varepsilon_b$):

$$\varepsilon_b = [(1 - \nu_o K_t)/(1 - K_t)] [\varepsilon'_b - K_t (\varepsilon'_a + \varepsilon'_c)/(1 + K_t)]$$

Solved example 3

- $K_t = 0.02$, $\nu_o = 0.285$; indicated $\varepsilon'_a = 1000\ \mu\varepsilon$ and $\varepsilon'_c = -300\ \mu\varepsilon$.

$$\text{factor} = (1 - 0.0057)/(1 - 0.0004) = 0.9947$$

$\varepsilon_a = 0.9947\ (1000 + 0.02 \times 300) = 1000.7\ \mu\varepsilon$ $\varepsilon_c = 0.9947\ (-300 - 0.02 \times 1000) = -318.3\ \mu\varepsilon$

- The error in the smaller strain ε_c is about 6 %, while ε_a has a negligible error. So correct whenever the transverse strain is large compared with the axial strain, or K_t is above 1 %.

Other corrections in rosette work

- Temperature (use self-temperature-compensated gauges and a dummy gauge), lead-wire resistance, non-linearity of bridge at large strains, finite size of the rosette (stacked rosettes reduce this), reinforcing effect of gauge.

5. Non-Destructive Testing (NDT) of Concrete – Introduction

- NDT finds strength, quality, cracks, voids and deterioration of concrete without damaging the structure.
- Why needed: doubtful cube results, old structures, fire/earthquake damage, quality control during construction, deciding formwork removal time.
- Main methods: rebound (Schmidt) hammer, ultrasonic pulse velocity (UPV), penetration resistance (Windsor probe), pull-out test, cover meter, half-cell potential (corrosion), radiography, core cutting (partially destructive).
- Standards: IS 13311 (Part 1) – UPV; IS 13311 (Part 2) – Rebound hammer; IS 516 (Part 5).

6. Ultrasonic Pulse Velocity (UPV) Method

6.1 Principle

- An ultrasonic pulse (frequency 24 – 150 kHz, usually 54 kHz) is sent through concrete from a transmitting transducer; a receiving transducer records the transit time T.

  Pulse velocity $V = L / T$ (L = path length, T = transit time)

- Sound travels faster in dense, good concrete than in porous, cracked concrete, so velocity indicates quality.
- The pulse velocity depends on the elastic constants and density:

  $$V = \sqrt{\{E(1-\mu)/[\rho(1+\mu)(1-2\mu)]\}} \rightarrow E_d = \rho V^2 (1+\mu)(1-2\mu)/(1-\mu)$$

6.2 Equipment

- Pulse generator, two piezo-electric transducers (T and R), amplifier, time-interval measuring circuit (resolution 0.1 μs), display; calibration (reference) bar of known time; couplant (grease / jelly) for good acoustic contact.

6.3 Arrangement of transducers

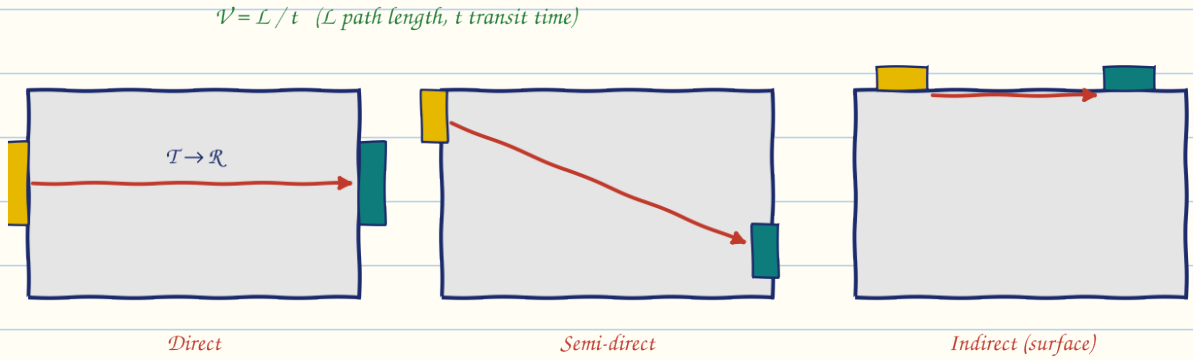


Fig 3.3 Direct, semi-direct and indirect transmission

- Direct (cross) transmission – transducers on opposite faces; most reliable, maximum pulse energy – preferred.
- Semi-direct – on adjacent faces; less sensitive, path length is measured centre-to-centre.
- Indirect (surface) – both transducers on the same face; weakest signal, velocity is 5–20 % lower; find V from the slope of the $T – L$ graph for several distances; used where only one face is accessible or to detect surface cracks.

- Minimum path length: 100 mm for aggregate ≤ 20 mm and 150 mm for aggregate ≤ 40 mm.

6.4 Procedure

- Set the reference-bar time; smooth the surface; apply couplant; hold the transducers firmly; measure L accurately; read T (take 3 readings); compute $V = L/T$.

6.5 Quality of concrete (IS 13311 Part 1)

- V above 4.5 km/s → Excellent; 3.5 – 4.5 → Good; 3.0 – 3.5 → Medium (doubtful); below 3.0 → Poor.

6.6 Factors affecting pulse velocity

- Moisture (wet concrete – higher V), aggregate type and quantity (more aggregate – higher V), w/c ratio (lower – higher V), age (increases with age), reinforcement (velocity along steel is higher – keep path away from bars, apply correction), cracks and voids (longer path – lower V), path length (> 100 mm), temperature (10–30 °C – negligible), stress level and transducer contact.

6.7 Applications of UPV to concrete

- Homogeneity (uniformity) check at many points; detect honeycombing, voids, cracks.
- Depth of surface-opening crack – transducers placed at equal distances x on either side of crack:

$$h = x\sqrt{[\,(T_c / T_s)^2 - 1\,]}$$ (T_c = time across crack, T_s = time over $2x$ on sound concrete)

- Strength estimation – V–f_c correlation graph made from cores / cubes of the same concrete; accuracy ± 15–25 %.
- Dynamic modulus of elasticity E_d; monitoring early-age setting, fire damage, freeze-thaw, sulphate attack, alkali-aggregate reaction.

Strength correlation with UPV (approximate guide)

- V = 3.5 km/s → about 15 – 20 N/mm²; V = 4.0 km/s → 25 – 35 N/mm²; V = 4.5 km/s → 40 – 50 N/mm². These vary with aggregate and mix, so a site-specific curve from cores or cubes must be drawn (log f_c vs V is usually a

straight line).

- *Procedure for the calibration curve:* cast cubes of different strength (change w/c); measure V through the cube before crushing, crush to get f_c, plot f_c against V and fit a curve.
- *Example:* $L = 300$ mm, $T = 71.4$ μs → $V = 300/71.4 = 4.2$ km/s (good). With $\rho = 2400$ kg/m³ and $\mu = 0.2$:

$$E_d = 2400 \times 4200^2 \times (1.2 \times 0.6/0.8) = 3.8\times10^{10}\ \text{Pa} = 38\ \text{GPa}$$

- *Crack-depth example:* $x = 150$ mm, $T_s = 70$ μs, $T_c = 98$ μs → $h = 150\sqrt{(1.4^2 - 1)} = 147$ mm.
- *Limitations:* needs calibration for strength, not directly strength; influenced by moisture and steel; cost of equipment.

7. Rebound (Schmidt) Hammer Test

7.1 Principle

- A *spring-loaded mass* strikes a steel plunger resting on the concrete surface. The *rebound distance* of the mass depends on the *surface hardness*, which is related to the *strength* of concrete. Rebound reading is the *rebound number* R (scale 10–100).
- Impact energy: 2.25 N·m (type N, ordinary concrete); 0.735 N·m for small members (type L).

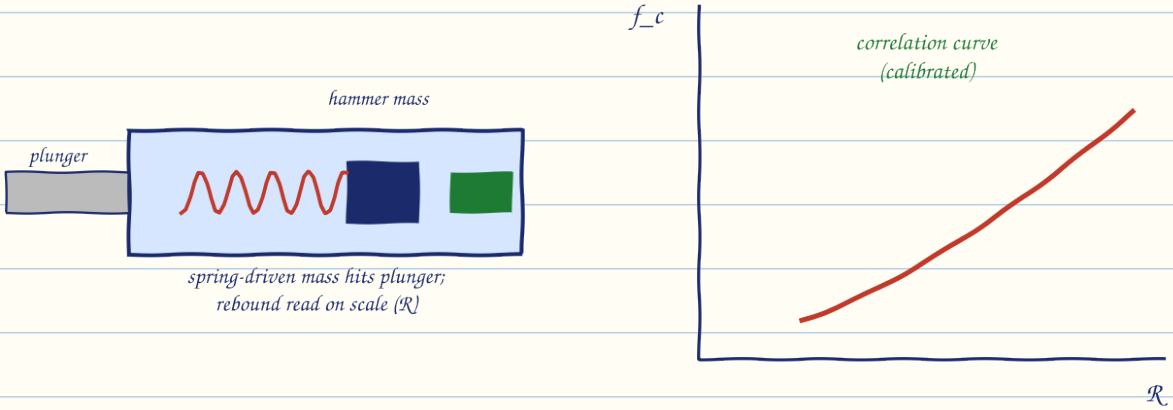


Fig 3.4 Rebound hammer and its correlation curve

7.2 Procedure (IS 13311 Part 2)

- Calibrate on the steel anvil supplied ($R \approx 80 \pm 2$).
- Choose a smooth, clean, dry, surface (grind rough patches; avoid honeycomb, rebar, edges within 20 mm).
- Hold the hammer perpendicular to the surface; push the plunger slowly; after impact note R.
- Take at least 9 (usually 12) readings on a 300 mm × 300 mm area, points ≥ 20 mm apart, discard readings deviating more than ±6 from the average and re-average.
- Apply the orientation correction (upward / downward / vertical) and use the correlation curve to obtain compressive strength.

7.3 Typical correlation (cubes, horizontal impact)

- $R \approx 20 \rightarrow$ about 10 – 15 N/mm^2; $R \approx 30 \rightarrow 20 – 25$ N/mm^2; $R \approx 40 \rightarrow 35 – 40$ N/mm^2; $R \approx 50 \rightarrow 55 – 60$ N/mm^2 (approximate – use the manufacturer's calibration chart).

7.4 Factors affecting rebound number

- Smoothness of surface; moisture (wet surface lowers R by up to 20 %); carbonation (raises R, remove the skin); age and curing; type of cement, aggregate and mix; size and stiffness of the member (thin members absorb impact); stress state, temperature, direction of impact.

7.5 Application to concrete

- Uniformity of concrete in a structure; locating weak zones; comparing the quality of different members; checking early strength (formwork removal); assessing old buildings; supplementary to core tests.
- Advantages: quick, cheap, light, simple, many readings in short time.
- Limitations: measures surface hardness (top 30 mm only), not true strength; accuracy ± 25 %; results depend on moisture, carbonation, aggregate; cannot replace cube or core tests.
- Combined method (SonReb): use R and V together in a multiple-regression formula; error reduces to about ± 10 %.

Solved example 4 – rebound readings

- Twelve readings at a point: 34, 36, 35, 33, 38, 34, 35, 36, 35, 27, 34, 35.

  Average of all = 412/12 = 34.3; 27 differs by 7.3 > 6 → reject

  Average of the remaining 11 = 385/11 = 35.0; all others lie within ±6 of 35 → $\mathcal{R}$ = 35

- From the calibration chart $\mathcal{R}$ = 35 → compressive strength ≈ 28 – 32 N/mm² (M30-type concrete).
- If the hammer pointed downward, the actual $\mathcal{R}$ would be corrected by about −3 (downward impact gives higher $\mathcal{R}$); for upward impact add about +3 (values from the chart supplied with the instrument).

Comparison

- UPV – looks through the depth, finds voids and cracks, quality grading, more costly. Rebound hammer – surface only, strength indicator, very cheap and quick. Both together give a reliable assessment.

8. Quick Revision – Unit III

- Rosette = 3 gauges at a point → $\varepsilon_1, \varepsilon_2, \theta$ (plane stress).

  Rectangular: $\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_c)/2 \pm (1/\sqrt{2})\sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2]}$, $\tan 2\theta = (2\varepsilon_b - \varepsilon_a - \varepsilon_c)/(\varepsilon_a - \varepsilon_c)$

  Delta: $\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_b + \varepsilon_c)/3 \pm (\sqrt{2}/3)\sqrt{[\Sigma(\text{differences})^2]}$, $\tan 2\theta = \sqrt{3}(\varepsilon_b - \varepsilon_c)/(2\varepsilon_a - \varepsilon_b - \varepsilon_c)$

- Transverse sensitivity: $\varepsilon_a = [(1 - \nu_0 K_t)/(1 - K_t^2)](\varepsilon'_a - K_t \varepsilon'_c)$.
- UPV: $V = L/T$; > 4.5 excellent, 3.5–4.5 good, 3–3.5 medium, < 3 poor.
- Rebound hammer: $\mathcal{R}$ number → strength by chart; surface hardness; ± 25 %.

Important questions

- What is a strain rosette? Why are three gauges required? Derive the principal strains for a rectangular rosette.
- Derive the expressions for principal strains and stresses for a delta rosette.
- What is transverse sensitivity? Derive the correction for a 0°/90° gauge combination.
- Explain the principle, procedure and applications of the UPV method for concrete.

- Describe the rebound hammer test. List the factors affecting the rebound number.
- Compare the UPV and rebound hammer methods of NDT.

Experimental Stress Analysis

UNIT – IV

THEORY OF PHOTOELASTICITY

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Introduction – Temporary Double Refraction – The Stress-Optic Law – Effects of Stressed Model in a Polariscope for Various Arrangements – Fringe Sharpening – Brewster's Stress-Optic Law.

Handwritten-style study notes · B.Tech Civil Engineering – IV Year

1. Introduction to Photoelasticity

- Photoelasticity is a whole-field optical method of stress analysis. A transparent model of a birefringent plastic (epoxy, polycarbonate) is loaded and viewed in polarised light; it shows coloured or dark fringes which give the stress distribution over the entire model.
- It is based on the property of temporary double refraction (birefringence) – a stressed transparent isotropic material behaves like an optical crystal whose axes lie along the principal stress directions.
- Landmarks: 1816 Brewster discovered stress-induced birefringence in glass; 1853 Maxwell gave the stress-optic law; 1930s Coker and Filon developed 2-D photoelasticity; Frocht, Mindlin, Post and Dally – modern developments.

What does photoelasticity give?

- Isochromatic fringes → difference of principal stresses ($\sigma_1 - \sigma_2$) everywhere.
- Isoclinic fringes → directions of the principal stresses everywhere.
- Free boundary: one principal stress is zero, so the boundary stress (where failure usually starts) is obtained directly with high accuracy.

Merits and limitations

- Merits: full-field picture, finds stress concentrations immediately, works for any shape and loading, good for teaching, cheap models for 2-D, extended to 3-D by stress freezing.
- Limitations: needs special model material and a polariscope, model-to-prototype scaling, only elastic stresses; time-edge effect and creep in plastics; separation of σ_1 and σ_2 needs extra work (Unit V); 3-D slicing is laborious.

Types

- 2-D transmission photoelasticity (plane models; Unit V), 3-D photoelasticity (stress freezing, slicing), reflection photoelasticity (a birefringent coating bonded to a real component), scattered-light photoelasticity.

2. Nature of Light and Polarisation

- Light is an electromagnetic transverse wave. The light vector (electric field E) vibrates perpendicular to the direction of travel.

$$c = f\lambda \quad n = c / v \quad E = a \sin[(2\pi/\lambda)(ct - z)] \quad \text{Intensity } I \propto a^2$$

- λ for visible light = 400 – 700 nm; sodium lamp 589 nm; mercury green 546 nm (used in laboratory).
- Refractive index n of a medium = speed of light in vacuum / speed in the medium.

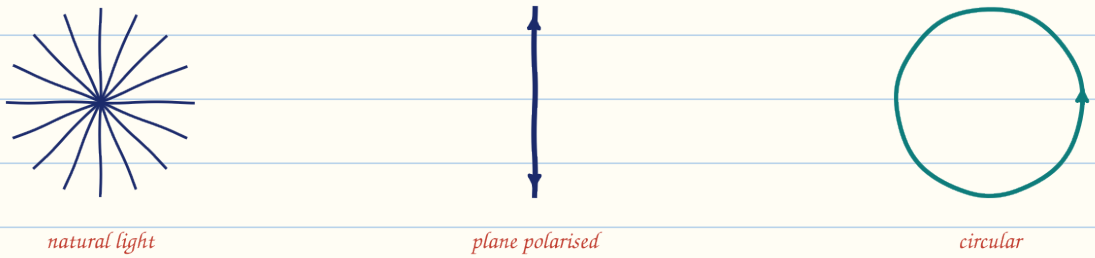


Fig 4.1 Natural, plane-polarised and circularly polarised light

- Natural (unpolarised) light: vibrations in all directions in a plane perpendicular to the ray.
- Plane (linearly) polarised light: vibration only in one plane.
- Circularly polarised light: two perpendicular components of equal amplitude with phase difference $\pi/2$ – tip of E traces a circle; elliptically polarised: unequal amplitude or phase.

Ways of obtaining polarised light

- Reflection at the polarising (Brewster) angle: $\tan\theta_B = n$; absorption (dichroism) in Polaroid sheet – most used in polariscope; double refraction (Nicol prism); scattering.
- Polaroid: a sheet of stretched polyvinyl alcohol with iodine – transmits only the vibration parallel to its transmission axis.

Malus' law

$$I = Io \cos^2\theta \quad (\theta = \text{angle between the polariser and analyser axes})$$

- $\theta = 90°$ (crossed) → $I = 0$ (dark field); $\theta = 0$ (parallel) → $I = Io$ (light field).

3. Double Refraction and Wave Plates

- In anisotropic crystals (calcite, quartz, mica) a ray splits into two:
  - Ordinary ray (O) – obeys Snell's law, speed same in all directions.

- Extraordinary ray (E) – does not obey Snell's law, speed depends on direction.

- Both rays are plane polarised at right angles to each other. The optic axis is the direction along which there is no double refraction.

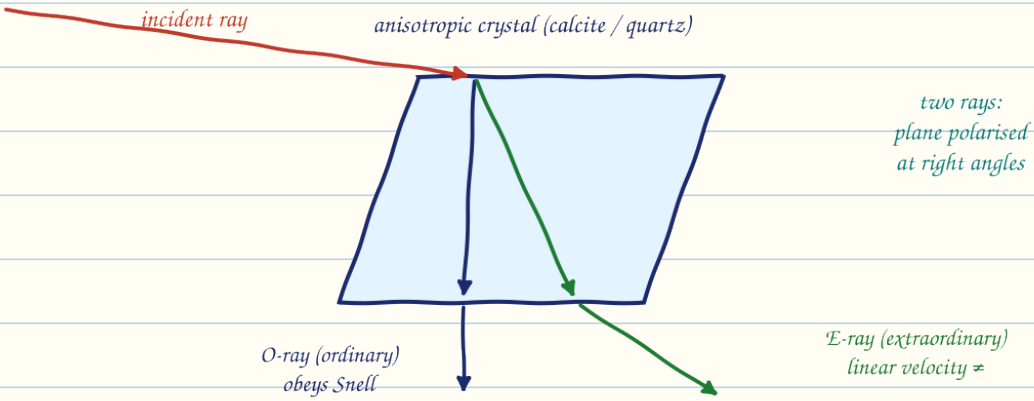


Fig 4.2 Double refraction in a crystal

- In a plate cut parallel to the optic axis the two rays travel along the same path with different speeds and emerge with a relative retardation:

$$\Delta = t\,(n_e - n_o) \text{ phase difference } \delta = 2\pi\Delta/\lambda$$

Wave plates

- Quarter-wave plate (QWP): $\Delta = \lambda/4$ ($\delta = \pi/2$). Plane polarised light at 45° to its axes becomes circularly polarised, and vice versa.
- Half-wave plate: $\Delta = \lambda/2$ ($\delta = \pi$). Rotates the plane of polarisation by 2θ.
- Thickness of a QWP: $t = \lambda/(4\,|n_e - n_o|)$. Quartz ($\Delta n \approx 0.0092$, $\lambda = 589$ nm): $t = 589\times10^{-9}/(4\times0.0092) \approx 16$ µm. Practical QWPs are of mica or plastic film, made for one wavelength.

Circularly polarised light – derivation

- Two components of equal amplitude a along x and y with a phase difference of $\pi/2$ (produced by a quarter-wave plate at 45°):

$$E_x = a\sin\omega t \quad E_y = a\sin(\omega t - \pi/2) = -a\cos\omega t \Rightarrow E_x^2 + E_y^2 = a^2$$

- This is the equation of a circle – the tip of the light vector rotates with constant amplitude; right or left circular depending on the sign of the phase difference.

- Plane polarised light at 45° to the axes of a QWP → circular; circular light through a second QWP (suitably set) → plane polarised again.

4. Temporary Double Refraction

- Many transparent, isotropic, non-crystalline materials (glass, celluloid, Bakelite, epoxy, polycarbonate, gelatin) are optically isotropic when free of stress.
- When stressed, they become doubly refracting like a crystal with optic axes in the directions of principal stresses σ_1 and σ_2; a ray entering the stressed plate splits into two plane polarised components vibrating along σ_1 and σ_2, travelling with different velocities.
- The effect lasts only while the load is applied and disappears on unloading – hence temporary double refraction.
- If the model is heated, loaded and cooled under load, the birefringence is permanently locked (stress freezing – used for 3-D photoelasticity).

Plane vs circular polarisation in the stressed model

- The size of retardation between the two components depends on thickness t and on $(\sigma_1 - \sigma_2)$ – this is the stress-optic law.

5. The Stress-Optic Law (Maxwell) and Brewster's Law

- For a 2-D stress state ($\sigma_3 = 0$) the refractive indices for light vibrating along the principal directions are linear functions of principal stresses:

$$n_1 - n_0 = C_1\sigma_1 + C_2\sigma_2$$
$$n_2 - n_0 = C_1\sigma_2 + C_2\sigma_1$$

- n_0 = refractive index of unstressed material; C_1, C_2 = absolute stress-optic coefficients (properties of material).
- Subtracting:

$$n_1 - n_2 = (C_1 - C_2)(\sigma_1 - \sigma_2) = C\,(\sigma_1 - \sigma_2)$$

- C = relative stress-optic coefficient, unit Brewster: 1 Brewster = 10^{-12} m^2/N.

Brewster's stress-optic law

- Relative retardation (optical path difference) for light passing normally through a model of thickness t:

$$\Delta = (n_1 - n_2)\,t = C\,t\,(\sigma_1 - \sigma_2)$$

- Statement: "The relative retardation developed between the two components is directly proportional to the thickness of the model and to the difference of the principal stresses."
- Number of wavelengths of retardation = fringe order N:

$$N = \Delta/\lambda = C\,t\,(\sigma_1 - \sigma_2)/\lambda \Rightarrow \sigma_1 - \sigma_2 = N\lambda/(C\,t) = N f_\sigma / t$$

- $f_\sigma = \lambda/C$ = material fringe value (N/m per fringe). It depends on material and on wavelength.
- Typical values (λ = 546 nm):
  - Epoxy (Araldite): $C \approx 50$ Brewster, $f_\sigma \approx 11$ kN/m per fringe.
  - Polycarbonate: $C \approx 78$ Brewster, $f_\sigma \approx 7$ kN/m per fringe (very sensitive).
  - PMMA (Perspex): $C \approx 4$–5 Brewster, $f_\sigma \approx 120$ kN/m (insensitive).

Solved examples

- Example 1. Epoxy model, t = 6 mm, f_σ = 11 kN/m. At a point N = 3 fringes.

$$\sigma_1 - \sigma_2 = N f_\sigma / t = 3 \times 11\,000 / 0.006 = 5.5\times10^6\ N/m^2 = 5.5\ MPa$$

- Example 2. Epoxy slab t = 5 mm, C = 50 Brewster, $\sigma_1 - \sigma_2$ = 4 MPa, λ = 546 nm.

$$\Delta = C\,t\,(\sigma_1 - \sigma_2) = 50\times10^{-12} \times 0.005 \times 4\times10^6 = 1.0\times10^{-6}\ m = 1000\ nm\ N = 1000/546 = 1.83$$

- Example 3 (free boundary). At a free edge $\sigma_2 = 0$, so $\sigma_1 = N f_\sigma/t$ directly: for N = 2, f_σ = 7 kN/m (polycarbonate), t = 5 mm → $\sigma_1 = 2 \times 7000/0.005 = 2.8$ MPa.

Remarks

- The law is valid only within the elastic limit and for small, instant loading (creep and the time-edge effect in plastics alter C).
- f_σ depends on wavelength (dispersion) – always state the light used. Use calibration to find f_σ for each model sheet.
- To increase sensitivity choose a material with low f_σ (high C) and a greater thickness.

6. Effect of a Stressed Model in a Polariscope

- Consider a plane polariscope: polariser P and analyser A crossed (axes at 90°), model between them. At a point the principal stress σ_1 makes angle α with the polariser axis.

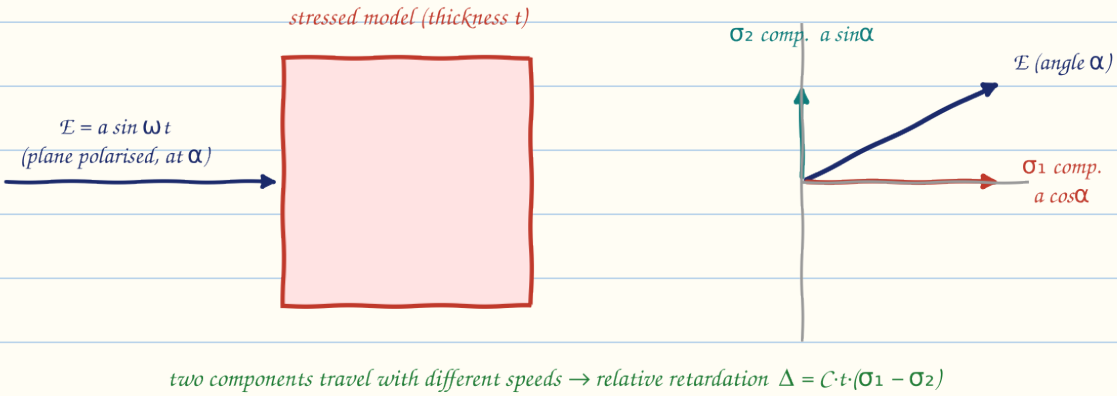


Fig 4.3 Resolution of light vector in a stressed model

- Light emerging from P: $E = a \sin \omega t$. In the model it splits into two components along σ_1 and σ_2:

$$E_1 = a \cos\alpha \sin\omega t \text{ (along } \sigma_1) \quad E_2 = a \sin\alpha \sin\omega t \text{ (along } \sigma_2)$$

- After passing through thickness t, the component with the higher index lags; with relative phase $\delta = 2\pi\Delta/\lambda = 2\pi N$:

$$E_1 = a \cos\alpha \sin\omega t \quad E_2 = a \sin\alpha \sin(\omega t - \delta)$$

- The analyser transmits only the components resolved along its own axis (perpendicular to P):

$$E_A = E_2 \cos\alpha - E_1 \sin\alpha = a \sin\alpha \cos\alpha \,[\sin(\omega t - \delta) - \sin\omega t]$$
$$E_A = -\, a \sin 2\alpha \cdot \sin(\delta/2) \cdot \cos(\omega t - \delta/2)$$

- Amplitude = $a \sin 2\alpha \sin(\delta/2)$, so the intensity of light that reaches the screen:

$$I = K a^2 \sin^2 2\alpha \cdot \sin^2(\pi \Delta/\lambda) = K a^2 \sin^2 2\alpha \cdot \sin^2(\pi N) \text{ (dark-field plane polariscope)}$$

Conditions of extinction (dark points)

- (i) $\sin 2\alpha = 0 \rightarrow \alpha = 0°$ or 90°: the principal stress direction coincides with the polariser (or analyser) axis → black band = isoclinic fringe (independent of

wavelength and of load).

- (ii) $\sin(\pi N) = 0 \rightarrow N = 0, 1, 2, 3, \ldots$: relative retardation is a whole number of wavelengths → dark isochromatic fringe (for monochromatic light).
- Maximum brightness when $\alpha = 45°$ and $N = ½, 3/2, 5/2 \ldots$

Light-field (polariser parallel to analyser)

$$I = K a^2 [\, 1 - \sin^2 2\alpha \cdot \sin^2(\pi N) \,]$$

- Bright where $N = 0, 1, 2, \ldots$; dark where $N = ½, 3/2, \ldots$. So dark-field gives integer orders, light-field gives half orders – both are used for fringe location.

White light

- Each wavelength is extinguished at a different N, so coloured isochromatics appear in the sequence black – grey – white – yellow – red – tint of passage (violet) – blue – green – yellow – red … $N = 0$ is black. Useful to identify the zero-order fringe and the sign of stress; monochromatic light is used for counting.

Sequence of colours with white light (tint of passage)

- Retardation Δ (nm): 0 – black; ≈ 280 – grey; ≈ 450 – white; ≈ 530 – yellow/orange; ≈ 565 – tint of passage (sensitive purple-violet); ≈ 600 – blue; ≈ 700 – blue-green; then the second-order series repeats with brighter colours.
- The tint of passage (retardation ≈ 1 fringe ≈ 575 nm) changes abruptly from red to blue – a sharp reference for fringe order 1. Beyond about the 3rd–4th order the colours wash out to pink-green.

7. Polariscope Arrangements

7.1 Plane (linear) polariscope

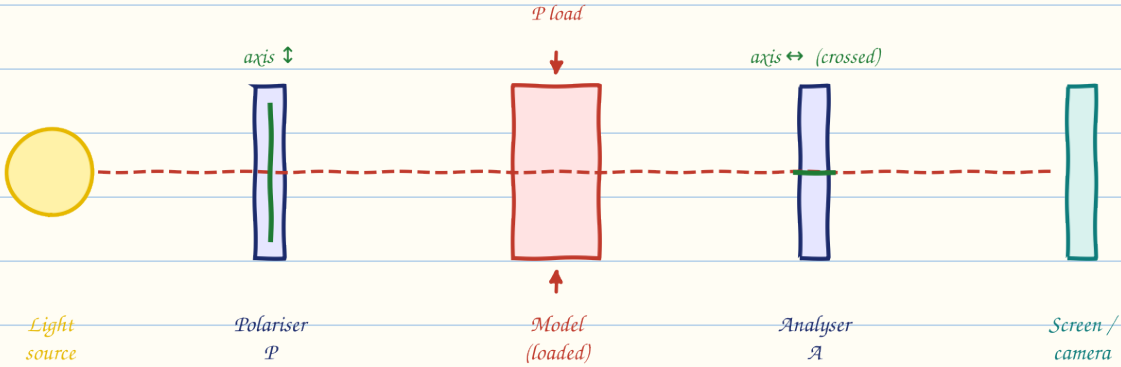


Fig 4.4 Plane polariscope (crossed arrangement)

- Components: light source (white or monochromatic), polariser P, loaded model M, analyser A, screen / camera.
- Crossed (dark field): axes of P and A perpendicular – $I = K a^2 \sin^2 2\alpha \sin^2 \pi N$; dark field when no model or no stress.
- Parallel (light field): axes parallel – bright field.
- A plane polariscope shows both isochromatics and isoclinics together (isoclinic black band confuses the picture).

7.2 Circular polariscope (adds two quarter-wave plates)

- Order: source → P → Q1 (QWP at 45° to P) → model → Q2 (QWP) → A → screen. Q1 converts plane polarised light into circularly polarised light, so the isoclinic dependence on α disappears and only isochromatics are visible.
- Four standard arrangements (P, A: polariser and analyser; Q1, Q2: quarter-wave plates):
- A: P ⊥ A, Q1 ⊥ Q2 (crossed) – dark field: $I = K a^2 \sin^2(\pi N)$
- B: P ⊥ A, Q1 ∥ Q2 (parallel) – light field: $I = K a^2 \cos^2(\pi N)$
- C: P ∥ A, Q1 ⊥ Q2 – light field: $I = K a^2 \cos^2(\pi N)$
- D: P ∥ A, Q1 ∥ Q2 – dark field: $I = K a^2 \sin^2(\pi N)$
- Procedure to choose arrangement: dark field (A or D) gives integer fringe orders ($N = 0, 1, 2 \ldots$); light field (B or C) gives half orders ($N = ½, 3/2 \ldots$). Combined use improves location of fringe centres.

Plane polariscope or circular polariscope?

- Isoclinics: use plane polariscope (rotate P and A together). Isochromatics: use circular polariscope. (Unit V)

Components of a practical polariscope

- Light source (high-pressure mercury vapour lamp with green filter 546 nm, or sodium lamp 589 nm; white tungsten lamp for colours) + diffuser.
- Field (collimating) lenses – produce a parallel beam; diameter decides the size of the model that can be viewed (e.g. 150 – 300 mm).
- Polariser, analyser and two quarter-wave plates in rotating mounts with graduated circles (to set angles).
- Loading frame (screw or dead-weight) with a load cell; model holder.
- Camera lens / screen / digital camera to record the fringe pattern.

Plane versus circular polariscope

- Plane: only P and A; shows isochromatics + isoclinics; used to find principal stress directions.
- Circular: P, Q1, Q2, A; shows only isochromatics; used to find $\sigma_1 - \sigma_2$.
- Plane polariscope needs no wave plates, so no wavelength error; the circular one is exactly correct only for the wavelength for which the QWPs were made.

Stress freezing (3-D photoelasticity) in brief

- Epoxy model heated slowly to the critical (stress-freezing) temperature (~ 120 °C) → loaded → cooled very slowly (2–3 °C/h) under load → load removed.
- The birefringence stays locked; the frozen fringe value is only about 0.3 – 0.5 kN/m (about 20–40 times more sensitive than at room temperature).
- The model is sliced (3 – 6 mm thick slices) and each slice is examined in a polariscope like a 2-D model.

8. Fringe Sharpening (and Multiplication)

- Problem: the intensity law $I \propto \sin^2(\pi N)$ is a smooth cosine curve, so the dark fringes are broad and their centres are difficult to locate, especially at fractional orders and when fringes crowd together.
- Fringe sharpening = reducing the width of the fringes so that the positions of integer (or half) orders can be located accurately. Fringe multiplication =

increasing the number of fringes in the model so that regions of low stress show measurable fringes.

Methods

- Monochromatic light (sodium or mercury) – black, well defined fringes instead of coloured bands.
- Post's method of fringe sharpening and multiplication – a series of partial mirrors (partially reflecting plates) placed before and after the model (parallel to it) at the polariser/analyser gives multiple passes of the light through the model. Intensity pattern becomes

  $I \propto \sin^2(\pi N) / [\, 1 + F \sin^2(\pi N)\,]$ (F depends on mirror reflectivity)

  - The result is very narrow bright (or dark) fringes and multiplied fringes ($N \to 2N, 3N, \ldots$) – analogous to the sharp fringes of a Fabry–Perot interferometer.
- Photographic methods – high-contrast (high gamma) film, double exposure, reducing background.
- Digital image processing – video camera, threshold and skeletonise (thin) the fringes; computer locates the fringe centres.
- Compensation techniques (Tardy, Babinet-Soleil) – measure fractional fringe order without sharpening (Unit V).

Why it matters

- Accurate fractional fringe order at the point of interest → accurate stress; essential at stress concentration (fillets and holes) where fringes crowd.

9. Quick Revision – Unit IV

- Photoelasticity = temporary double refraction + polarised light → whole-field stresses.

Stress-optic law: $n_1 - n_2 = C(\sigma_1 - \sigma_2)$; $\Delta = C\,t\,(\sigma_1 - \sigma_2)$; $\sigma_1 - \sigma_2 = N f_\sigma / t$; $f_\sigma = \lambda / C$

Plane polariscope: $I = K a^2 \sin^2 2\alpha \sin^2 \pi N$ (isoclinics from $\sin 2\alpha$, isochromatics from $\sin \pi N$)

- Circular polariscope: 4 arrangements, $I = K a^2 \sin^2 \pi N$ (dark) or $\cos^2 \pi N$ (light) – no isoclinics.
- Fringe sharpening: monochromatic light, Post's partial mirrors, digital processing.
- Units: Brewster = 10^{-12} m^2/N; fringe value in N/m per fringe.

Important questions

- What is temporary double refraction? Explain with a neat sketch.
- State and derive the stress-optic law. Define Brewster, fringe value and fringe order.
- Derive the expression for the intensity of light emerging from an analyser in a plane polariscope with a stressed model. Explain isoclinics and isochromatics.
- Describe the arrangements of a circular polariscope with intensity equations.
- What is fringe sharpening? Explain Post's method.
- A model of thickness 6 mm with f_σ = 11 kN/m shows N = 4; find $\sigma_1 - \sigma_2$. (Ans: 7.33 MPa)

Experimental Stress Analysis

UNIT – V

TWO DIMENSIONAL PHOTOELASTICITY

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Introduction – Isochromatic Fringe Patterns – Isoclinic Fringe Patterns – Passage of Light through Plane Polariscope and Circular Polariscope – Compensation Techniques – Calibration Methods – Separation Methods – Scaling Model to Prototype Stresses – Materials for Photoelasticity – Properties of Photoelastic Materials.

Handwritten-style study notes · B.Tech Civil Engineering – IV Year

1. Introduction – Two-Dimensional Photoelasticity

- 2-D photoelasticity deals with thin flat models (plane stress) loaded in their own plane; polarised light passes normally through the thickness.
- Stress state at every point is described by σ_1, σ_2, θ ($\sigma_3 = 0$). The polariscope gives:
  - Isochromatic fringes → $\sigma_1 - \sigma_2$ (principal stress difference).
  - Isoclinic fringes → θ (principal stress direction).
  - Separation methods (additional work) → $\sigma_1 + \sigma_2$, hence σ_1 and σ_2 separately.
- Models: thin epoxy, polycarbonate plates, 3 – 10 mm thick, cut to the shape of the structure (plate with hole, gear tooth, crane hook, dam, beam, bracket, tunnel section).
- Advantage of a 2-D model: one model, one loading – the stress over the whole model is seen at once, and boundary stresses (where cracks start) are got directly.

Steps of a 2-D photoelastic analysis

- Prepare the model → calibrate the material (find f_σ) → mount in the polariscope.
- Load and record isochromatics (circular polariscope, dark and light field).
- Record isoclinics (plane polariscope) and draw stress trajectories.
- Find fractional fringe order by compensation; separate σ_1 and σ_2; scale to the prototype.

2. Isochromatic Fringe Patterns

- Definition: an isochromatic is the locus of points with the same relative retardation – i.e. the same difference of principal stresses ($\sigma_1 - \sigma_2$). With monochromatic light they are dark bands (dark field); with white light coloured bands.

$$\sigma_1 - \sigma_2 = N f_\sigma / t \quad (N = 0, 1, 2, 3 \ldots \text{ on the dark fringes})$$

- Each fringe is a contour line of the maximum shear stress $\tau_{max} = (\sigma_1 - \sigma_2)/2 = N f_\sigma/(2t)$.

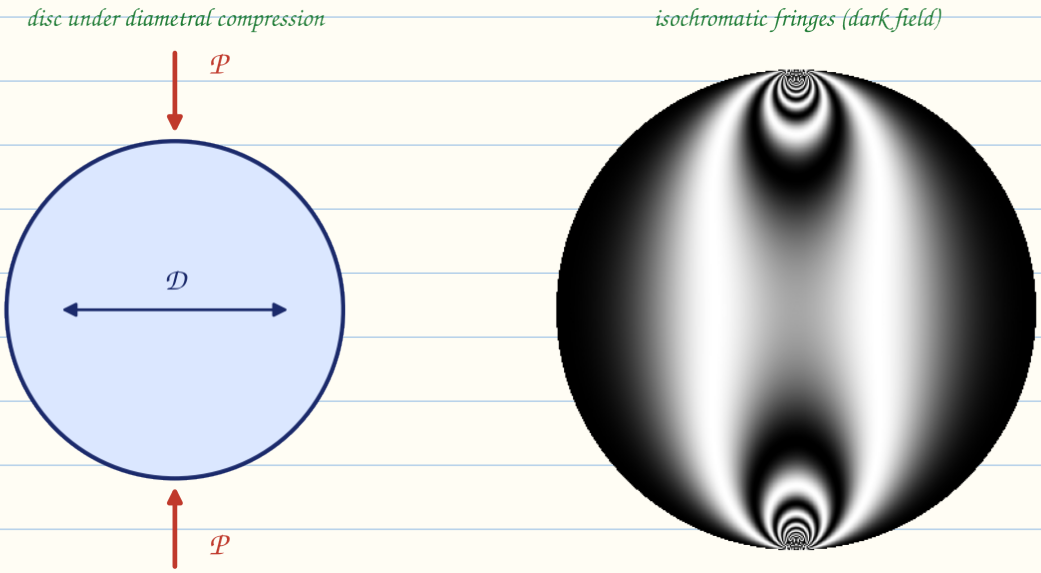


Fig 5.1 Disc under diametral compression and its isochromatic fringes

Properties of isochromatics

- $N = 0$ (black in white light) occurs where $\sigma_1 = \sigma_2$ (isotropic point) or both are zero; fringes never cross each other (a point cannot have two values of N).
- Closely spaced fringes → high stress gradient (near holes, fillets, loading points); widely spaced fringes → uniform stress.
- On a free boundary one principal stress is zero, so the boundary stress is $\sigma_b = N f_\sigma / t$, directly.
- Fringe order increases with load (new fringes are born at the points of highest stress and move toward lower-stress regions).
- Fringes are independent of the orientation of the polariser (circular polariscope) but depend on wavelength (position of colours).
- In the diametral-compression disc the highest N is at the load points; at the centre N is given by $8P/(\pi D f_\sigma)$.

Counting the fringe order

- Start from a point of known $N = 0$ (a free corner, a lightly stressed region, or the isotropic point; the black fringe in white light) and count outward; watch fringes as the load is slowly increased from zero.
- Use white light: the tint of passage marks the 1st fringe; use monochromatic light for accurate positions.

3. Isoclinic Fringe Patterns

- Definition: an isoclinic is the locus of points at which the principal stress directions are parallel to the axes of the polariser and analyser. It is a black band whose parameter θ = angle of polariser axis.

From $I = K a^2 \sin^2 2\alpha \sin^2 \pi N$: extinction when $\sin 2\alpha = 0 \rightarrow \alpha = 0°, 90°$

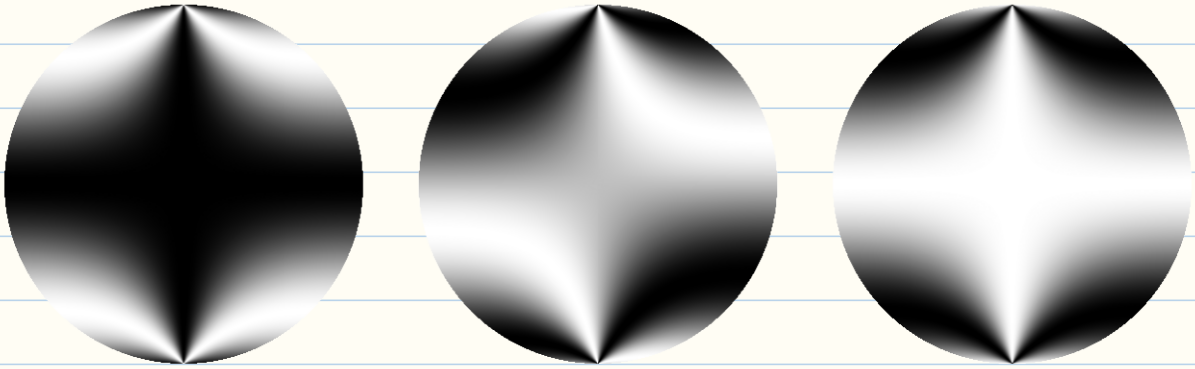


Fig 5.2 Isoclinic fringes of the disc for polariser angles 0°, 30°, 45°

How to obtain isoclinics

- Use the plane polariscope (crossed polariser and analyser; no quarter-wave plates).
- Rotate polariser and analyser together in steps of 5° or 10° (0°→90°); at each setting trace the black band on tracing paper and label it with the angle θ.
- Isoclinics are black in white light and monochromatic light alike and do not move when the load is changed (only their intensity changes) – this distinguishes them from isochromatics.
- Use low load (so few isochromatics) and a white light source to see isoclinics clearly.

Properties

- Along an axis of symmetry the principal directions are along and perpendicular to the axis → the 0° / 90° isoclinic lies on that axis.
- At free boundary one principal direction is tangent to the boundary.
- Isotropic point ($\sigma_1 = \sigma_2$): all isoclinics pass through it (direction indeterminate).
- Singular point: a point where $\sigma_1 = \sigma_2 = 0$ – also all isoclinics meet.

Uses of isoclinic data

- Stress trajectories (isostatics): lines whose tangent at every point is a principal stress direction; draw by joining the isoclinic angles – two orthogonal families (σ_1 and σ_2 trajectories). They show the flow of force and help to place reinforcement (e.g. in concrete members).
- Direction θ is required for separation of stresses (shear difference method).

4. Passage of Light through the Plane and Circular Polariscope

4.1 Plane polariscope (summary of Unit IV)

- Polariser gives $E = a \sin \omega t$. In the model it splits along σ_1 ($a \cos\alpha \sin \omega t$) and σ_2 ($a \sin\alpha \sin \omega t$), relative phase $\delta = 2\pi N$. Analyser (crossed) passes

$$I = K a^2 \sin^2 2\alpha \cdot \sin^2(\pi N) \text{ (dark field) } I = K a^2 [1 - \sin^2 2\alpha \sin^2(\pi N)] \text{ (light field)}$$

- Both isoclinic ($\sin^2 2\alpha$) and isochromatic ($\sin^2 \pi N$) effects appear.

4.2 Circular polariscope

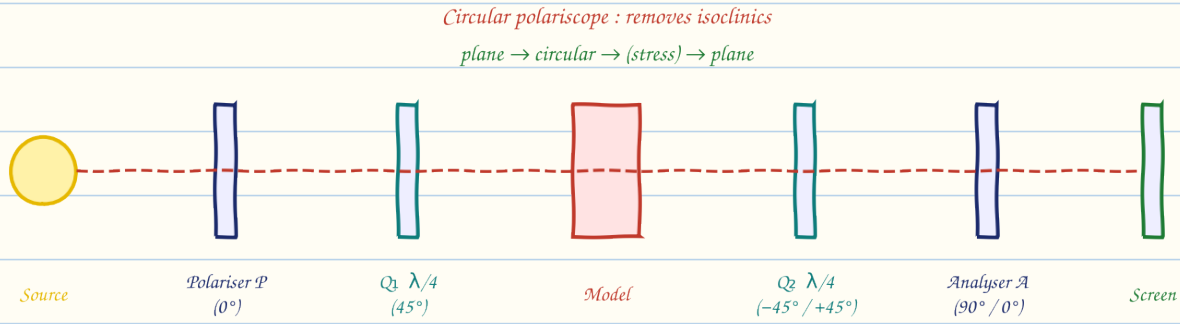


Fig 5.3 Circular polariscope: P, Q_1, model, Q_2, A

- Arrangement: P (0°) → Q_1 (fast axis at 45°) → model → Q_2 (axis at 45° ⊥ or ∥ to Q_1) → A.
- After P: $E = a \sin \omega t$. After Q_1 (QWP at 45°): two components of amplitude $a/\sqrt{2}$ with phase $\pi/2$ → circularly polarised light:

$$E_f = (a/\sqrt{2}) \sin \omega t \; E_s = (a/\sqrt{2}) \sin(\omega t - \pi/2) = -(a/\sqrt{2}) \cos \omega t$$

- The circular light contains equal components along any pair of perpendicular axes – so the orientation α of σ_1 drops out. After the model (phase $\delta = 2\pi N$ between σ_1, σ_2 components), through Q_2 and A (crossed arrangement A), the emerging

amplitude is $a \sin(\delta/2) = a \sin(\pi N)$:

$$I = K a^2 \sin^2(\pi N) \text{ (dark field – circular polariscope); } I = K a^2 \cos^2(\pi N) \text{ (light field)}$$

- Result: no isoclinics – only isochromatics (integer orders in dark field, half orders in light field).
- Dark field (crossed P-A, crossed Q1-Q2): $N = 0, 1, 2\ldots$; light field (parallel Q1-Q2 with crossed P-A): $N = ½, 3/2, 5/2\ldots$ – used together to locate the fringe positions more exactly.

Plane or circular? – remember

- Plane polariscope → isoclinics (and isochromatics mixed); circular polariscope → isochromatics only.

5. Compensation Techniques (Fractional Fringe Order)

- Problem: the point where stress is wanted rarely lies on a fringe centre; its order is $N = n +$ fraction. Compensation gives the fractional part.

5.1 Tardy method of compensation (most used)

- Equipment: plane polariscope + quarter-wave plate + analyser rotation scale.
- Step 1: In the plane polariscope rotate P and A together until the isoclinic passes through the point of interest – the polariser is now parallel to σ_1 (or σ_2) at that point.
- Step 2: Turn the analyser back to the cross setting, insert Q2 (QWP) with its axis along the polariser axis, and Q1 is not used. (i.e. polariser at the principal direction, QWP at 45°…) – The light leaving the model is elliptical.
- Step 3: Rotate the analyser by an angle β until the nearest lower integer fringe n moves to the point of interest (extinguishes the point).

$$N = n + \beta / 180° \ (\beta \text{ in degrees; fractional order} = \beta/180°)$$

- Example: point lies between the 2nd and 3rd fringes. The analyser rotation needed to bring the 2nd fringe to the point is 54°.

$$N = 2 + 54/180 = 2.30$$

- Advantages: only a QWP and rotation scale needed; accuracy ± 0.02 fringe; used in laboratory.

5.2 Babinet and Babinet–Soleil compensator

- Two quartz wedges with optic axes at right angles, one fixed and one moved by a micrometer screw; the Soleil modification has a parallel plate for a uniform field. They give a known, adjustable retardation.
- Place the compensator at the point of interest with its axes parallel to σ_1, σ_2; adjust until the model fringe at the point is cancelled; the compensator reading is the retardation of the model at that point (calibrated in nm or fringe order).

5.3 Senarmont and other methods

- Senarmont: quarter-wave plate + rotating analyser (same as Tardy but QWP axis is parallel to σ_1).
- Fringe counting at increasing load, fringe interpolation/multiplication, white-light tint of passage, and digital phase-shifting methods also give fractional orders.

Why compensation is important

- Error of 0.1 fringe with f_σ = 11 kN/m and t = 6 mm → error in $(\sigma_1 - \sigma_2)$ = 0.1×11 000/0.006 = 0.18 MPa. Needed for accuracy at stress-concentration points.

6. Calibration Methods (Finding the Material Fringe Value f_σ)

- The fringe value of each batch of sheet is found by loading a simple specimen of the same material, thickness and age whose stress is known by theory.

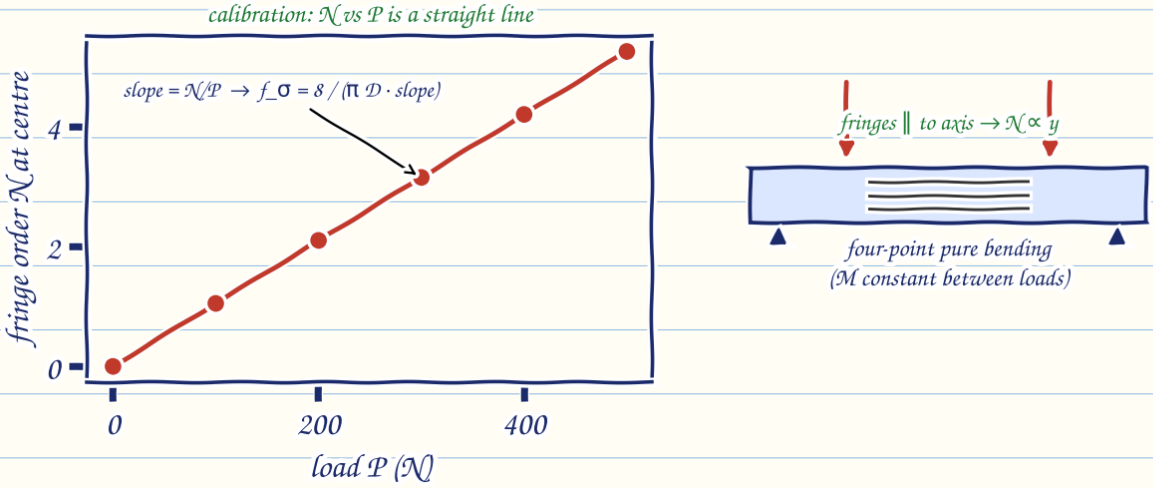


Fig 5.4 Calibration: N vs P is linear; four-point bending beam

6.1 Tension specimen

- A strip of width b and thickness t carries load P: $\sigma_1 = P/(b\,t)$, $\sigma_2 = 0$.

$$f_\sigma = \sigma_1\, t / N = P / (b\, N)$$

- Example: $b = 25$ mm, $P = 1650$ N, $N = 6.0 \rightarrow f_\sigma = 1650/(0.025 \times 6) = 11\,000$ N/m = 11 kN/m.

6.2 Pure bending (four-point) beam

- $\sigma_1 = M\,y / I$ (varies linearly), N varies linearly with depth y; at the extreme fibre $\sigma_{max} = 6M/(b\,h^2)$.

$$f_\sigma = \sigma_{max}\, t / N_{max} = 6\,M\,t / (b\,h^2\,N_{max})$$ (with $b = t$ for a rectangular section: $f_\sigma = 6M/(h^2\,N_{max})$)

- Fringes are straight lines parallel to the neutral axis – easy to count; also checks the linearity of the material.

6.3 Circular disc under diametral compression (best and commonest)

- At the centre of a disc of diameter D and thickness t under a diametral load P:

$$\sigma_x = +\,2P/(\pi\,D\,t)$$ (tension, horizontal) $$\sigma_y = -\,6P/(\pi\,D\,t)$$ (compression, vertical)

$$\sigma_1 - \sigma_2 = 8P/(\pi\,D\,t) \Rightarrow f_\sigma = 8P / (\pi\,D\,N)$$

- Increase P in steps, count N at the centre each time and plot N vs P → straight line; f_σ from the slope.
- Example: $D = 50$ mm, $P = 1000$ N, $N = 4.6 \rightarrow f_\sigma = 8\times1000/(\pi\times0.05\times4.6) = 11\,070$ N/m ≈ 11.1 kN/m.
- Advantages: only a compression load is required; the centre point is easy to observe; simple model.

Precautions in calibration

- Same sheet, thickness, temperature, wavelength, and time after loading as the actual model; load in small increments and measure N after the fringe pattern has stabilised (1–2 min).

7. Separation Methods ($\sigma_1 + \sigma_2$ and individual stresses)

- Photoelasticity gives $(\sigma_1 - \sigma_2)$ and θ – two quantities; three $(\sigma_1, \sigma_2, \theta)$ are required, so an extra relation is needed.

- *Free boundary:* one principal stress = 0 → σ at the boundary is obtained *without* separation.

7.1 Shear-difference method

- Uses the *equilibrium equation* (no body force):

$$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y = 0 \Rightarrow \sigma_x(B) = \sigma_x(A) - \int (\partial\tau_{xy}/\partial y)\, dx \approx \sigma_x(A) - \Sigma\, (\Delta\tau_{xy}/\Delta y)\,\Delta x$$

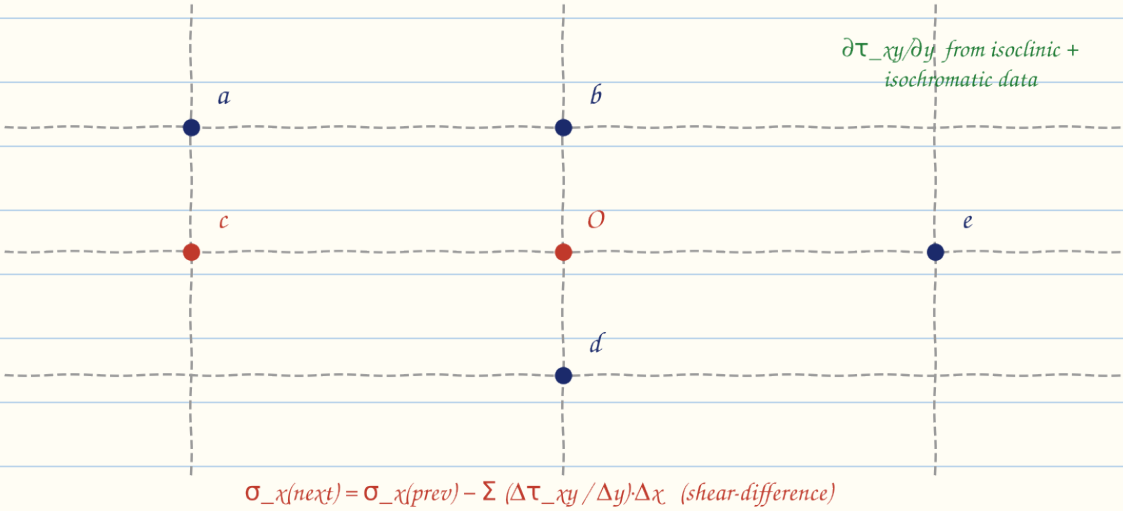


Fig 5.5 Shear-difference method: a grid along a line of known σ_x

- *From photoelastic data at any point:*

$$\tau_{xy} = [(\sigma_1 - \sigma_2)/2] \sin 2\theta = (N f_\sigma / 2t) \sin 2\theta \quad \sigma_x - \sigma_y = (\sigma_1 - \sigma_2) \cos 2\theta$$

- *Procedure:* (i) choose a line (usually a *free boundary or an axis where* σ_x *is known*); (ii) draw *parallel lines at* $\pm\Delta y$ and a series of points at spacing Δx; (iii) find N and θ at each point (isochromatic + isoclinic data), hence τ_{xy} above and below; (iv) $\sigma_x(\text{next}) = \sigma_x(\text{prev}) - (\Delta\tau/\Delta y)\Delta x$; (v) $\sigma_y = \sigma_x - (\sigma_1 - \sigma_2)\cos 2\theta$ and $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm (\sigma_1 - \sigma_2)/2$.
- *Example (step):* Δx = 5 mm; τ_{xy} above = 2.6 MPa, below = 3.8 MPa, spacing $2\Delta y$ = 4 mm:

$$\partial\tau_{xy}/\partial y \approx (2.6 - 3.8)/4 = -0.3 \text{ MPa/mm}; \quad \Delta\sigma_x = -(-0.3)(5) = +1.5 \text{ MPa}$$

- *Solved example – at a point:* N = 3, f_σ = 11 kN/m, t = 6 mm, θ = 30°, σ_x = 4 MPa.

$\sigma_1 - \sigma_2 = 3\times11000/0.006 = 5.5$ MPa; $\sigma_y = \sigma_x - 5.5\cos 60° = 4 - 2.75 = 1.25$ MPa

$\tau_{xy} = 2.75 \sin 60° = 2.38$ MPa; $\sigma_{1,2} = 2.625 \pm 2.75 \rightarrow \sigma_1 = 5.375$ MPa, $\sigma_2 = -0.125$ MPa

7.2 Other separation methods

- *Laplace equation method (numerical):* for plane stress without body force, $\nabla^2(\sigma_1 + \sigma_2) = 0$ (harmonic). Boundary values of $(\sigma_1 + \sigma_2)$ come from the free boundary; interior values by *finite-difference/relaxation*, or by *electrical-analogy* and *membrane-analogy* models.
- *Oblique incidence (Drucker):* tilt the model by angle φ about the principal axis and note the change of fringe order → gives σ_1 and σ_2 separately.
- *Lateral strain measurement:* thickness change $\Delta t = -(\nu\, t / E)(\sigma_1 + \sigma_2)$ measured with *Mach–Zehnder interferometer* or a *lateral extensometer* → $\sigma_1 + \sigma_2$ directly.
- *Analytical/Numerical:* Airy stress function, FEM using photoelastic boundary data.

8. Scaling Model to Prototype Stresses

- In *plane stress* (simply connected body, no body force) the stress distribution depends only on *geometry and the loading*, *not on E*, and only weakly on ν. If the model is geometrically similar in-plane:

$$\sigma \propto P / (L\, t)$$

$$\sigma_p = \sigma_m \cdot (P_p / P_m) \cdot (L_m / L_p) \cdot (t_m / t_p)$$

- For a *distributed pressure* p on a boundary the stress scales directly: $\sigma_p/\sigma_m = p_p/p_m$.
- *Example:* model stress $\sigma_m = 5.5$ MPa under $P_m = 600$ N. Prototype: $P_p = 24$ kN, model/prototype length ratio $L_m/L_p = 1/4$, thickness $t_m/t_p = 6/20 = 0.3$.

$$\sigma_p = 5.5 \times (24\,000/600) \times (1/4) \times 0.3 = 5.5 \times 40 \times 0.25 \times 0.3 = 16.5 \text{ MPa}$$

- *Choice of model load:* choose P_m so that there are *8 – 12 fringes* at the critical region (good accuracy) without exceeding the *proportional limit* ($\sigma \approx 15 – 20$ MPa) or creep.

- Strains and deflections scale as σ/E: $\varepsilon_p = \varepsilon_m (\sigma_p/\sigma_m)(E_m/E_p)$.
- If $\nu_m \neq \nu_p$ and the body has holes (multiply connected), apply a small correction for Poisson's ratio.

9. Materials for Photoelasticity

- Essential requirements of a photoelastic material
  - Transparent and optically isotropic when unstressed (free of residual stress, homogeneous).
  - High optical sensitivity (low f_σ, high C) – many fringes at low stress.
  - Linear elastic (σ–ε and N–σ linear) up to a high proportional limit; high E (small deformation, good for accuracy).
  - No creep and no time-edge effect (stable fringe pattern).
  - Easy to cast and machine without heating or introducing stress; stable with time, humidity, temperature.
  - Low cost; available in sheets of uniform thickness.
- Figure of merit (sensitivity) $Q = E / f_\sigma$ – larger is better.

Commonly used materials (approximate values at room temperature)

- Epoxy resin (Araldite, CT-200, PLM-4): $E \approx 3$ GPa, $\nu \approx 0.36$, $f_\sigma \approx 11$ kN/m, low creep; best for 2-D and 3-D (stress freezing) models.
- Polycarbonate (PSM-1, Lexan): $E \approx 2.5$ GPa, $\nu \approx 0.38$, $f_\sigma \approx 7$ kN/m – very sensitive; slight creep; heats during machining; used for 2-D and reflection coatings.
- CR-39 (allyl diglycol carbonate): $E \approx 2.6$ GPa, $f_\sigma \approx 26$ kN/m; good optical quality.
- Bakelite / phenolic resin: $f_\sigma \approx 10$ kN/m; large time-edge effect; early 3-D work.
- PMMA (Perspex / Plexiglas): $E \approx 3$ GPa, $f_\sigma \approx 120$ kN/m – insensitive, cheap, used for demonstration models only.
- Celluloid, glass: older materials; glass used in 3-D scattered-light work.
- Urethane rubber, gelatin: very low E (few MPa), $f_\sigma \approx 0.2$ kN/m – used for large deformation, dynamic and low-load models.

Properties of photoelastic materials – what to check

- Fringe value f_σ and stress-optic coefficient C (calibrate each sheet), modulus E, Poisson's ratio ν, proportional limit, creep and time-edge effect, residual (initial) stress, temperature sensitivity, homogeneity, machinability, optical clarity.
- Time-edge effect: an apparent change in birefringence at the edges of a model with time (moisture absorption / machining stress) – reduce by coating edges with wax, using sharp cutters at low speed, annealing and reading the model soon after loading (within minutes).

Model preparation and loading

- Cast or purchase sheets; cut with fine band-saw or laser, mill with sharp tools at low speed and coolant, file and polish; anneal to remove residual stress.
- Apply load through pins, rollers and soft pads to avoid local friction; keep model in the loading frame free of bending (out-of-plane).

10. Quick Revision – Unit V

- Isochromatic: $\sigma_1 - \sigma_2 = N f_\sigma / t$; contour of τ_{max}; dark in circular polariscope; $N = 0$ isotropic point.
- Isoclinic: direction of principal stress = polariser axis; plane polariscope, rotate P-A together; gives stress trajectories.
- Plane: $I = K a^2 \sin^2 2\alpha \sin^2 \pi N$; circular: $I = K a^2 \sin^2 \pi N$ (dark) or $\cos^2 \pi N$ (light).
- Compensation: Tardy – $N = n + \beta/180°$; Babinet–Soleil compensator.
- Calibration: disc $f_\sigma = 8P/(\pi D N)$; tension $f_\sigma = P/(b N)$; beam $f_\sigma = 6Mt/(b h^2 N)$.
- Separation: shear-difference, Laplace equation, oblique incidence, lateral strain; $\sigma_x - \sigma_y = (\sigma_1 - \sigma_2)\cos 2\theta$; $\tau_{xy} = ((\sigma_1 - \sigma_2)/2) \sin 2\theta$.

Scaling: $\sigma_p = \sigma_m (P_p/P_m)(L_m/L_p)(t_m/t_p)$

- Materials: epoxy (11 kN/m), polycarbonate (7 kN/m), CR-39, PMMA (120 kN/m), urethane rubber.

Important questions

- Distinguish isochromatic and isoclinic fringes. How are they obtained and what information do they give?

- Explain the passage of light through a plane polariscope and a circular polariscope; why are isoclinics absent in the latter?
- Describe the Tardy method of compensation with a neat sketch. Find N if n = 3 and $\beta = 72°$. (Ans: 3.40)
- Describe the calibration of a photoelastic material by the circular disc method; derive $f_\sigma = 8P/(\pi D N)$.
- Explain the shear-difference method of separating the principal stresses.
- How are model stresses scaled to the prototype in 2-D photoelasticity?
- List the requirements of a good photoelastic material; compare epoxy, polycarbonate and PMMA.